

Formalizing Algebraic Quantum Field Theory with Deep Learning Theorem Proving in Lean

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Abstract

Deep learning has gone from success to success, e.g. from AlphaGo defeating world champion Go player Lee Sedol to AlphaFold's Nobel Prize in Chemistry. With DeepSeek-Prover-V2, it's also made significant mathematical inroads. However, there's been no similar progress for physics; it's not been used to *prove* results in physics. With the increasing axiomization of physics, e.g. Haag-Kastler axioms, and the rising prevalence of proof assistants, e.g. Lean, this, however, will soon change. The work proposed here will underpin that change. It will: (1) create, train, and maintain an open MCTS DeepSeek-Prover-V2 based deep learning model for theorem proving in algebraic quantum field theory; (2) integrate that model into Lean as an open interactive proof assistant (IPA); and (3) use that IPA to formalize, and open source, the foundational axioms and theorems of algebraic quantum field theory.

Keywords: Formalization, Algebraic Quantum Field Theory, Deep Learning, Lean

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1 Introduction

1.1 Why Now?

Within just the last few years, progress in deep learning has opened up possibilities previously thought impossible:

- Lee Sedol, a Go player who ranked second all-time in international titles [163], lost a 5 game tournament 1-4 against AlphaGo [127], a deep learning model.
- OpenAI and GitHub introduced GitHub Copilot [104], a deep learning model that is able to act as a pair programmer to help developers write better code [104].
- DeepSeek-AI introduced and open sourced DeepSeek-Prover-V1.5 and DeepSeek-Prover-V2, deep learning models able to prove mathematical theorems [171, 119].

And this isn't even mentioning ChatGPT [107]. Even Terence Tao, a 2006 Fields Medalist [123], has come to believe [144] that deep learning based proof assistants can help us to start “doing mathematics on a scale which is really unprecedented”.

While there has been much success utilizing deep learning for mathematics [119, 171, 85, 153, 116, 118], similar progress has not been made for the field of physics. Though computers have long been used to simulate physical phenomena [120], they have not been widely applied to prove theorems in physics. However, with physics moving towards a more mathematical foundation [65, 63, 5], deep learning becoming ever more powerful, and proof assistants like Lean [101] becoming more prevalent, this will soon change.

1.2 Objectives

The work proposed here endeavors to be the foundational step in this change. It will:

- Create, train, and maintain a MCTS DeepSeek-Prover-V2 based deep learning model for theorem proving in algebraic quantum field theory.
- Integrate this deep learning model into the Lean environment as an interactive proof assistant (IPA).
- Use this interactive proof assistant (IPA) to formalize the foundations of algebraic quantum field theory in Lean.
- Open source the model, integration, and formalization while also promoting these to maximize impact and kick-start further research progress.

2 Literature Review

2.1 Theorem Proving with Deep Learning

Since the 1950's computers have been utilized to prove mathematical theorems [44]; at the time, this amounted to “mechanically” applying deductive rules to reach logical conclusions. “Learning” based approaches [140] only appeared in the 90's, much later. It's only since about 2016 that deep learning approaches [76, 84] have become prevalent, and it's only very recently [49, 170, 171] that deep learning approaches have had successes approaching the level of an undergraduate math major [11].

In this section we review these deep learning approaches, briefly touching upon the main threads of research and the techniques used.

2.1.1 Background

Theorem proving in mathematics has a long history, going all the way back to at least Euclid’s Elements [68] some 2300 years ago. The method of proof used since then, logical argument in a natural language, has recently become known [93] as “informal theorem proving”. It’s what most mathematicians engage in on a daily basis. However, despite mathematicians’ best efforts, informal theorem proving always opens up the possibility for mistakes. Mathematicians are indeed human.

More recently, in an effort to eliminate such mistakes, mathematicians and computer scientists invented “formal theorem proving”. This involves representing theorems and proofs in a machine verifiable format, guaranteeing¹ their correctness [93].

2.1.2 Methods and Tasks

Deep learning has ushered in new methods and tasks for theorem proving. In addition, it has enhanced old methods and tasks, e.g. automating things which heretofore were done by-hand [93]. Here we briefly touch upon four examples: autoformalization, premise selection, proof step generation, and proof generation.

2.1.2.1 Autoformalization Autoformalization is the process of a machine automatically translating an informal proof, a natural language proof, into a formal proof, a proof in a machine verifiable format.

Broadly stated, the process of autoformalization is that of converting a sequence of characters, the informal proof, to a second sequence of characters, the formal proof. Hence, as one might expect, sequence-to-sequence models, like those used in machine translation [139, 33], excel at this task [155, 156].

As one might also expect, there are models [169, 11, 81] trained for the inverse process informalization, going from a formal proof to an informal proof. Informalization is useful as it allows humans to quickly “grok” formal proofs and can be used to publish informal versions of computer generated formal proofs.

2.1.2.2 Premise Selection Mathematicians when building proofs have much more than just “raw” axioms in their toolbox. They also have the thousands of theorems, postulates, lemmas, and corollaries mathematicians before them have proved. For example, they may only need to use an existing lemma to complete a proof they’re working on. This process of selecting the proper theorem, postulate, lemma, or corollary is called premise selection.

Premise selection can be understood as a conditioned prediction problem, given a partial proof predict the next premise to use, i.e. predict the next theorem, postulate, lemma, or corollary to use. When selecting from a reasonable number of premises, one can simply use a model with a final logit layer to indicate which premise is favored; the initial work in premise selection did exactly this [76, 84].

As is generally the case these days, the recent successes [107] of LLM’s motivated researchers to use them, in particular BERT models [48], for premise selection [51, 161]. Follow-on research [174, 175, 42] simply moved onto using more powerful LLM’s for the same task.

2.1.2.3 Proof Step Generation Proof step generation is the process of generating the next step in a proof. It can also be understood as a conditioned prediction problem, given a partial proof predict the next step. Generally this next step is a tactic (e.g. a Lean command such as `refine` or `rw`) along with the tactic’s argument(s).

This framing of the problem leads to a fruitful research thread [162, 74, 13, 111, 122, 167, 121]. This thread of research uses two deep learning models for proof step generation. One model predicts the next tactic and a second model the tactic’s argument(s).

¹Assuming, of course, the computer’s kernel is correctly implemented.

As one might expect with the recent successes [107] of LLM’s, they have also been utilized in this task [116, 117, 66, 80, 177, 175, 173, 160, 151, 60, 54] with some success. This involves training an LLM to complete a sequence-to-sequence prediction task in which the input sequence is a partial proof, and the target sequence is the proof’s next step.

2.1.2.4 Proof Generation Proof generation can also be understood as a conditioned prediction problem, but instead of predicting only the next step as in proof step generation, proof generation predicts all subsequent steps, a problem exponentially more difficult.

The bulk of current proof generation research employs supervised learning, training on existing proofs to produce models capable of generating complete proofs [98, 34, 77, 9, 78, 137, 35, 61, 136, 53, 10, 62, 79, 14]. However, the exponential increase in difficulty relative to proof step generation forces the use of more advanced techniques.

A promising thread of research in this regard builds upon the success of AlphaGo [127, 128], utilizing Monte Carlo tree search (MCTS) [40] variants [85, 181, 153, 171] to boost the performance of a base model. As we will see in subsequent sections, this MCTS research thread seems far from exhausted and holds much promise.

2.1.3 Datasets

The Achilles heel of deep learning is training data. Obtaining enough high-quality training data is often the hardest part of the problem. Deep learning for theorem proving is no exception in this regard.

However, deep learning for theorem proving has the added complexity of requiring extremely high-quality data, data that only experts, e.g. mathematics professors, can produce. Such data is far more rare than say the “internet sized” datasets [21] required² for GPT-3.

In this section we will look at how researchers are attempting to solve the difficult problem of obtaining enough high-quality training data to train deep learning, theorem proving models. We will do so by examining the two major techniques used: data collection, i.e. extracting training data from existing sources, and data generation, i.e. generating training data ex nihilo.

2.1.3.1 Data Collection The easiest location to collect and extract high-quality data is from existing repositories of high-quality data, and initially this is exactly what researchers did. Communities built up around proof assistants such as Lean [101] created libraries such as `mathlib` [99] that contain thousands of formally proved theorems. So researchers set out to collect and aggregate such data into training datasets.

For example, LeanStep [66], LeanDojo [174], and MLFMF [15] all extract data from `mathlib` [99]. While these datasets are high-quality, they are by modern standards small. LeanDojo has about 98k theorems and proofs with about 130k premises, a large dataset from a human perspective, but small compared to the terabyte sized datasets LLM’s are commonly being trained on [21].

Another technique researchers are using to create high-quality datasets is to manually formalize informal proofs. As one might expect, such efforts are time-consuming and require highly skilled workers for the formalization process. Hence, again, the resulting datasets are not large.

In this vein miniF2F [179] manually formalizes 488 hard, high school level problems. FIMO [95] and ProofNet [11] manually formalize undergraduate level and International Mathematical Olympiad level problems in Lean. While these datasets are high-quality, they are unfortunately small.

To counterbalance the lack of large-scale datasets, some researchers have broadened their search criteria. Instead of focusing purely on existing, formal datasets or manually formalizing data, they

²CommonCrawl was 60% of the training data and the filtered CommonCrawl dataset was 570GB. So it was trained on 950GB of text data [21].

consider a larger category of “mathematical texts”, which can include things like webpages dedicated to mathematical topics. Such datasets aren’t targeted at training models, but only at pre-training, i.e. getting the model parameters conditioned so they may maximally benefit from the higher-quality, smaller datasets such as ProofNet [11]. Datasets in this vein include InfIMM-WebMath-40B [67], WebMath [66], Proof-Pile [11], and InternLM-Math [176].

2.1.3.2 Data Generation As large, high-quality datasets are non-existent and creating them manually takes far too much time and money, researchers starting exploring methods to automatically generate large, high-quality datasets.

Mathematics’ structure is conducive to automatic data generation. Provable statements can be generated by starting with a set of axioms then mechanically applying inference rules. This process is simple enough that it can be used programmatically to generate large datasets of theorems and their proofs. A research thread [167, 168, 53, 149, 2, 157] utilizing this process has been relatively fruitful in generating larger datasets.

An alternative to generating a theorem and its proof ex nihilo is to generate different proofs for a fixed set of theorems, e.g. the set of theorems from mathlib [99]. This has the advantage of always creating proofs of “significant” theorems, but the disadvantage of creating dataset with less diversity. (Dataset diversity helps the final trained model in dealing with a larger variety of theorems and proofs.)

Some of the earliest work in this regard [13, 116, 117] employs expert iteration [4], training a proof generation model on a dataset of theorems and their proofs while augmenting the dataset with the novel proofs the model generates in the process of training. More recent systems such as Gödel-Prover [94] have also employed this technique. In addition MCTS variants can also be viewed as work in this thread, the policy being trained on proofs created by the full MCTS system.

Another means of data generation is to simply “cache” partial results found during proof generation training and use those partial results in future proof (step) generation. Mathematicians do this all the time, “caching” partial results in theorems, propositions, lemmas, and corollaries. The two main works in this thread are LEGO-Prover [172] and REFACTOR [180].

2.1.4 Research Questions

The core research question is: *Can deep learning models match and then surpass human-level, theorem proving abilities?*

We have already seen deep learning models exhibit superhuman performance in the games of Go, Chess, and Shogi [129]. So the goal may be achievable. But, current deep learning models [49, 170, 171] only approach the level of an undergraduate math major, e.g. only solving 25% of the undergraduate level problems in ProofNet [11].

The only way to quantitatively track progress is through benchmarks, for example ProofNet and FrontierMath³ [59]. These benchmarks currently indicate we’re not there yet, but they also indicate progress is being made.

Here, at a high-level, we sketch research questions related to datasets and proof generation as they relate to the core research question above. In later sections we’ll dive into the details of all these questions.

2.1.4.1 Datasets When one compares the size of existing formal proof training datasets, e.g. the 98k theorems and proofs of LeanDojo [174], to the size of training datasets required for current LLM’s, e.g. the 950GB of GPT-3 [21], it’s clear that there is a dearth of formal proof training data for LLM’s.

³FrontierMath is a research level benchmark containing problems that would take researchers hours to days to complete.

Similarly, when one compares the rate at which formal proof training data can be manually created, a handful of theorems a day, to the size of training datasets required for current LLM's, it's also clear that one can't manually create formal proof training data fast enough.

Both of these suggest that the way forward is data generation, generating training data *ex nihilo*. The research question is: *How does one generate formal proof training data at volume while still maintaining a high level of quality?*

2.1.4.2 Proof Generation Of the many different algorithms applied to proof generation, MCTS variants seem the most promising.

For example, consider the results of a recent SotA model DeepSeek-Prover-V1.5, a MCTS variant [171]. On the miniF2F test set [179] the policy in isolation solved 60.2% of the problems while when the policy was used as part of the MCTS variant, the entire system solved 63.5% of the miniF2F test set problems. An improvement, but less than what one has come to expect from MCTS.

For example, the Go playing model AlphaGo Zero [128] utilized a MCTS variant where the policy in isolation had an Elo⁴ of 3055. However, when the policy was used as part of the MCTS variant, the entire system had an Elo of 5185. This means [41] the MCTS variant would be expected to lose or draw about 1 out of every 211350 games against the policy. The MCTS variant is far, far stronger.

This 60.2% vs 63.5% result for DeepSeek-Prover-V1.5 versus the 1 out of every 211350 games result for AlphaGo Zero strongly suggests that MCTS based theorem proving systems are far from their limit. This suggests the research question: *What MCTS variants improve performance on the most challenging benchmarks?*

This disparity between the DeepSeek-Prover-V1.5 results and the AlphaGo Zero results further suggests that MCTS based systems can play a key role in elevating deep learning theorem proving from an undergraduate level, to a graduate level, to that of a professor, to a superhuman level, a progression MCTS has already achieved in Go, Chess, and Shogi [129].

2.2 Formalizing Mathematics in Lean

Mathematics is the foundation upon which all other sciences are built. Physics, chemistry, and biology all rely upon the certain foundation provided by mathematics. This certainty is ensured by informal proofs, logical arguments in a natural language arguing for the validity of particular propositions. However, despite mathematicians' dogged efforts, sometimes errors creep into these informal proofs. Formalizing these proofs, putting them in a machine verifiable format, can eliminate such errors and thus further solidify the foundation of mathematics.

Formalizing mathematics has other benefits too. Deep learning models are becoming evermore powerful—they can synthesize speech [29]; play Go at a superhuman level [129]; and engage in human like discussions [107]. Generally, what makes this possible is training data. So, formalization, when viewed as creating training data, is one of the first steps in priming future deep learning models to reach superhuman levels in mathematics.

Also, formalization will likely provide benefits we can't even foresee [24]. When CD's first came to market, they were touted as a better sounding alternative to vinyl, the fact that the audio was digitized was secondary. However, with hindsight we now know the important change wasn't CD's themselves, which are now antiquated, but digitization and the new possibilities (e.g. streaming, audio manipulation, digital audio workstations) it provides. Formalization is the digitization of mathematics, and we don't yet know all the benefits it will bring.

⁴Elo [41] is a rating system for two player games of skill. It is used to determine how likely it is for one player to lose or draw against a second player.

2.2.1 Formalizing Mathematics in Lean

Without a doubt, ground zero of Lean based formalization is occurring in `mathlib` [99]—a community-led, free, open source library containing thousands of theorems and proofs. It already covers a broad swatch of mathematics. Topics include the basics (e.g. real numbers and complex numbers), more advanced topics (e.g. schemes and the Radon–Nikodym theorem), and some of mathematics’ forefront (e.g. perfectoid spaces [25]).

Though formalization always ends with commits to `mathlib` in GitHub, there are variety of different paths that lead to this destination. For example, undergrads as part of a student project may formalize something they’re learning about, e.g. Sylow’s theorems, conformal maps, and parts of Galois theory were all added via this route [25]. There are also longterm, formal projects that tackle more advanced topics, e.g. p-adic numbers [92], schemes [27], and perfectoid spaces [26]. Also, there are more informal additions, where someone just feels like `mathlib` needs “X” so they rollup their sleeves and do it, e.g. Morrison and Topaz developed a huge amount of category theory this way [25].

The key takeaway is that formalization in Lean is moving quickly and picking up steam. At the time of this writing and with Lean community’s rate of progress [25], many topics are covered at a PhD level or above.

2.2.2 Formalizing Physics in Lean

In contrast to the vibrant community formalizing mathematics in Lean, there is a vanishingly small community formalizing physics in Lean. The only significant effort in this direction is the HepLean [148] project which produced the PhysLean [71] library.

Unfortunately, PhysLean is sparse and contradictory. For example, it “formalizes” anomaly cancellation in QED [39, 1], but doesn’t formalize QED [52]. It “formalizes” quantum mechanics [71], but doesn’t formalize the axioms of quantum mechanics [65]. As a result, PhysLean tacitly assumes far too much to serve as a foundation upon which to confidently build.

As the examples above show, there are several problems with PhysLean. However, it seems that there are two central problems from which all others flow:

- **Lack of Foundations** - PhysLean does not layout the foundational axioms upon which it builds. For quantum mechanics it doesn’t formalize the axioms of quantum mechanics [65]. For quantum field theory it doesn’t formalize the Haag-Kastler axioms [63] or say the Wightman axioms [133].
- **False Fields of Focus** - PhysLean focuses on fields of physics which do not lend themselves to Lean’s strength, guaranteeing theorems and proofs are correct. PhysLean focuses on fields of physics where theorems and proofs play little to no role, e.g. Electromagnetism.

2.2.3 Research Questions

This current state of the PhysLean library raises the research questions: *Is it possible and/or useful to formalize physics in Lean?*

While PhysLean’s lack of foundations and false fields of focus make it appear that formalizing physics in Lean isn’t possible and/or useful, this isn’t the case. There are fields of physics, e.g. quantum mechanics [65] and quantum field theory, that have axiomatic frameworks [63, 133] where theorems and proofs are commonplace [63, 133, 65]. In these fields, formalization can bear the same fruits seen in mathematics and also benefit from the accelerated progress deep learning will doubtlessly bring.

In the context of the core research question above, we’ll focus here only on quantum field theory, sketching research questions related to its formalization. Later we’ll look into the details of these questions.

2.2.3.1 Formalizing Quantum Field Theory The overarching research questions related to quantum field theory are: *Is it possible and/or useful to formalize quantum field theory in Lean?*

To the “possible” question, it seems that it is indeed possible. There are various axiomatic frameworks for quantum field theory [63, 133]. However, the various axiomatic frameworks are obviously very different⁵. So, there’s the added question of which axiomatic framework to work with. Furthermore, even within a given framework, e.g the Haag-Kastler axioms, there are often slightly different sets of axioms put forward by different authors. Are all of these variants equivalent? If not, which should be built upon and why? All of these are open research questions.

To the “useful” question, the short answer is “definitely yes”. Formalization will bring quantum field theory’s axioms, theorems, postulates, and lemmas into a computable format. Deep learning models will then be able to learn from them, reestablish standard results and, as their reasoning abilities doubtlessly increase, derive new results, things we likely could not have foreseen.

However, as evidenced by the various axiomatic frameworks and their variants, the foundations of quantum field theory are still relatively uncertain. Even in research within one axiomatic framework, it’s often unclear what variant of the framework is being used⁶. Formalization and deep learning can help rectify this situation. Deep learning based interactive proof assistants (IPA) can accelerate the process of determining what framework variant is being used, quickly determining, say, if a theorem’s proof requires a particular axiom. Similarly, they can assist in formalizing the theorems, propositions, and lemmas within a framework.

2.3 Algebraic Quantum Field Theory

Quantum field theory is the most accurate physical theory ever constructed; experiment and theory agree to one part in a trillion [50]. Despite this, quantum field theory is built on a “shaky”⁷ mathematical foundation [143], putting physics in a quite a strange state of affairs.

Realizing this, physicists in the 1950’s attempted to remedy this situation. As Hilbert proposed in his sixth problem [72],

*...diejenigen physikalischen Disziplinen axiomatisch zu behandeln, in denen schon heute die Mathematik eine hervorragende Rolle spielt*⁸,

physicists began axiomizing quantum field theory, introducing the Wightman axioms [133], the Osterwalder-Schrader axioms [108], and the Haag-Kastler axioms [64]. All of which provide partial solutions to Hilbert’s sixth problem.

Here we discuss axiomatization’s history with a focus on algebraic quantum field theory’s Haag-Kastler axioms. We also discuss axiomatization’s many successes, e.g. CPT theorem, spin-statistics theorem, proof of the quantum null-energy condition, as well as many others.

2.3.1 Pre-Axiomatization

General relativity grew out of one deep, physical insight, the equivalence principle. From this everything follows, resulting in a minimal, coherent theory.

Quantum field theory, on the other-hand, grew out of the need to reconcile two contradictory theories, quantum mechanics and special relativity. As such, it grew organically, vestigial elements surgically attached to those critical to its survival⁹. This produced a theory that worked, but no one was quite sure

⁵For example the Wightman axioms are obviously very different from the Haag-Kastler axioms.

⁶For example, are the original Haag-Kastler axioms being used or the revision that appears later in Haag’s book?

⁷The introduction in *What is a Quantum Field Theory* [143] by Talagrand, a 2024 Abel Prize recipient [106], gives a refreshingly candid opinion of physics’ level of rigor.

⁸Translated: “...to treat those physical disciplines in which mathematics already plays a prominent role axiomatically”

⁹For example, in curved spacetime high energy particle physics is not a theory of particles but is a theory of fields [152].

why it worked, what its one, fundamental principal was. It was a theory with a coccyx and a heart, but with no clear way to distinguish the vestigial one.

2.3.2 Dawn of Axiomatization

With this as backdrop, the 1950's saw the birth of axiomatization. Many early efforts of this period—Heisenberg S-matrix program [69], Bogolyubov-Medvedev-Polivanov S-matrix approach [17], Lehmann-Symanzik-Zimmermann axioms [90]—have been all but forgotten [45], and understandably so. At the time delineating between what was essential and what was only “limbs and outward flourishes” was nigh impossible.

In contrast, though also birthed around this time, the Wightman axioms [133], Osterwalder-Schrader axioms [108, 110], and Haag-Kastler axioms [64] have all stood the test of time and still have an influence on modern physics. The Haag-Kastler axioms, in particular, are the foundation of algebraic quantum field theory.

Here we'll briefly cover these last three axiomatic systems, giving the flavor of each, their successes, and a bit on how they relate to one another.

2.3.2.1 Wightman Axioms The Wightman axioms [133] were the first attempt to meet with some success. A set of seven axioms [63] they formalize quantum field theory on Minkowski spacetime. Covering them formally will take us too far afield. However, informally they cover (1) the Hilbert space and Poincaré group, (2) the definition of fields, (3) the existence of a field's Hermitian conjugate, (4) fields' transformation properties under the Poincaré group, (5) causal commutation relations, (6) “completeness”¹⁰, and (7) “primitive causality”¹¹.

One can accomplish quite a lot with only these axioms. For example, the CPT theorem—which proves quantum field theory on Minkowski spacetime is invariant under the combination of a charge conjugation, a parity transform, and a time reversal—is a direct consequence [83]. The spin-statistics theorem—why Bosons commute and Fermions anti-commute—is also a direct result [133]. The surprising Reeh-Schlieder theorem¹² also follows from the Wightman axioms [63]. These deep results were a strong signal that the Wightman axioms were “on the right track”.

2.3.2.2 Osterwalder-Schrader Axioms The Osterwalder-Schrader axioms [109] are very similar to the Wightman axioms. Roughly the Osterwalder-Schrader axioms can be viewed as a Euclidian version of the Wightman axioms. In fact, the Osterwalder-Schrader theorem [109] proves that under certain conditions, theories described by the Osterwalder-Schrader axioms may be Wick rotated to theories described by the Wightman axioms.

2.3.3 Birth of Algebraic Quantum Field Theory

Motivated by Segal's championing of an algebraic approach [125] and by the success of the Wightman axioms, Haag and Kastler [64] proposed the Haag-Kastler axioms, the foundation of algebraic quantum field theory.

In contrast to the Wightman axioms, the Haag-Kastler axioms take a “purely algebraic” form, jettisoning the Wightman axioms' one, preferred vacuum. To one familiar with more traditional formulations of quantum field theory, this seems disastrous. However, it's quite the opposite; it's prescient.

¹⁰Roughly the statement that algebra of “smeared field operators” is dense in the space of bounded operators on the Hilbert space.

¹¹Basically the statement that there exists a dynamical law that allows one to compute the evolution of a field.

¹²The Reeh-Schlieder theorem roughly states that for any open bounded region $\mathcal{O}$ its algebra of “smeared field operators” $\mathcal{A}(\mathcal{O})$ when acting on the vacuum Ω is dense in the Hilbert space $\mathcal{H}$, i.e. $\mathcal{A}(\mathcal{O})\Omega$ is dense in $\mathcal{H}$.

For example, the Unruh effect [56, 43, 150], discovered years later, makes it clear that vacua are observer dependent. It shows that in Minkowski spacetime the vacuum of an inertial observer differs from the vacuum of a uniformly accelerating observer. To the uniformly accelerating observer, the inertial observer's vacuum appears to contain a thermal bath of particles. So there is no one, preferred vacuum. Prescience.

2.3.3.1 Haag-Kastler Axioms The six¹³ original Haag-Kastler axioms [64] formalize quantum field theory on Minkowski spacetime. As they are technical, covering them here in detail will take us too far afield. However, they can be briefly stated as follows:

Let $\{\mathcal{O}_i\}$ be the set of all open sets of Minkowski spacetime with compact closure, then the Haag-Kastler axioms state:

1. **Existence of a Net** - There exists an assignment, called a net, of an abstract C^* -algebra $\mathcal{C}(\mathcal{O}_i)$ to every $\mathcal{O}_i$ in $\{\mathcal{O}_i\}$.
2. **Isotony** - If $\mathcal{O}_1$ and $\mathcal{O}_2$ in $\{\mathcal{O}_i\}$ satisfy $\mathcal{O}_1 \subset \mathcal{O}_2$, then their abstract C^* -algebras satisfy $\mathcal{C}(\mathcal{O}_1) \subset \mathcal{C}(\mathcal{O}_2)$, where $\mathcal{C}(\mathcal{O}_1)$ and $\mathcal{C}(\mathcal{O}_2)$ have either a common unit or both have no unit.
3. **Local Causality** - If $\mathcal{O}_1$ and $\mathcal{O}_2$ in $\{\mathcal{O}_i\}$ are spacelike separated, then $\mathcal{C}(\mathcal{O}_1)$ and $\mathcal{C}(\mathcal{O}_2)$ commute.
4. **Existence of a Quasi-Local Algebra** - The set theoretic union of all $\mathcal{C}(\mathcal{O}_i)$ over $\{\mathcal{O}_i\}$ is a normed $*$ -algebra and its completion $\mathcal{C}$, called the quasi-local algebra, is a C^* -algebra.
5. **Existence of a Poincaré Representation** - There exists a representation α of the Poincaré group P on $\mathcal{C}$ such that if $p \in P$, then $\alpha_p(\mathcal{C}(\mathcal{O}_i)) = \mathcal{C}(p(\mathcal{O}_i))$ where $p(\mathcal{O}_i)$ is the image of $\mathcal{O}_i$ under the Poincaré transformation p .
6. **Existence of a Faithful, Irreducible Representation** - The C^* -algebra $\mathcal{C}$ has at least one faithful, irreducible representation.

As one can see, these are abstract and purely algebraic. There is no reference to a Hilbert space, a vacuum, or any other construct that one is accustomed to in quantum field theory. Interestingly, one finds that Hilbert spaces and vacua do appear but as *derived* concepts, quite different from what one is used to.

2.3.3.2 Successes of Haag-Kastler From these axioms one is able to derive deep results, a signal that the Haag-Kastler axioms are “on the right track” too. We briefly touch on some of these results below.

First, using what's come to be known as the GNS construction [58, 124], one is able to associate many different Hilbert spaces and different vacua to a single theory satisfying the Haag-Kastler axioms. This predated the discovery of the Unruh effect [56, 43, 150] in which different vacua are associated to a single theory. Also, vacua plural are impossible using the Wightman axioms. Wightman's first axiom states that there is only a single Hilbert space containing a single vacuum.

Another triumph of the Haag-Kastler axioms is a deeper understanding of superselection sectors. Using the GNS construction [63], any state—a normalized, positive linear form ω on $\mathcal{C}$ —defines a Hilbert space $\mathcal{H}_\omega$ and a representation π_ω of $\mathcal{C}$ on $\mathcal{H}_\omega$. Studying representations defined by “physically admissible”¹⁴ states, the Haag-Kastler axioms allow one to learn a great deal about generic properties

¹³The original article [64] also included a seventh ill-defined “axiom” on page 852. It stated that if $\mathcal{O}_1$ is in the “casual shadow” of $\mathcal{O}_2$, then their abstract C^* -algebras satisfy $\mathcal{C}(\mathcal{O}_1) \subseteq \mathcal{C}(\mathcal{O}_2)$. The term “casual shadow” was never defined. In later versions of the axioms, this seems to have been replaced [63] by $\mathcal{C}(\mathcal{O}) = \mathcal{C}(\mathcal{O}'')$, where $\mathcal{O}''$ is the “casual completion”, defined in [63], of $\mathcal{O}$.

¹⁴There are various definitions of “physically admissible” used here. Common ones are the DHR selection criterion [63] and the Borchers' selection criterion [63].

of superselection sectors in quantum field theory. For example, one is able to clarify the origin of Bose and Fermi statistics, the existence of global gage groups that describe particle charges, and many other results [22].

Yet another triumph of the Haag-Kastler axioms is a generalization of Goldstone's theorem [23]. This generalization allows for a larger class of symmetries than traditional gauge symmetries [158, 159]. In addition, its mechanism for symmetry breaking is distinct from the Higgs mechanism [158, 159]. Both of these are significant, but unfortunately underappreciated, results.

2.3.4 Renaissance of Tomita-Takesaki Theory

About five years after Haag and Kastler introduced their axioms, Minoru Tomita introduced what we now know as Tomita-Takesaki theory [147]. This started a renaissance in algebraic quantum field theory. However, Tomita's work was ignored when first introduced.

Tomita's work was ignored as it was unpublished and the initial preprint was riddled with errors [145]. Takesaki, instead of ignoring the work as most others had done, tried to rescue its core ideas [145]. This resulted in a complete revision and clarification of Tomita's work which Takesaki published [142] in 1970. It was then that algebraic quantum field theory was reborn.

2.3.4.1 Tomita-Takesaki Theory Describing Tomita-Takesaki theory in detail will take us too far afield; however, its core ideas [22, 138] can be presented in a few, brief paragraphs.

Consider a von Neumann algebra $\mathcal{N}$ and its Hilbert space $\mathcal{H}$. If Ω is a cyclic¹⁵, separating¹⁶ vector in $\mathcal{H}$, one can define an antilinear operator S on $\mathcal{N}\Omega$ by $S\mathcal{N}\Omega := \mathcal{N}^*\Omega$, for all elements N in $\mathcal{N}$. It follows that S is densely defined and closable. Hence, its closure $\bar{S}$ admits a polar decomposition $\bar{S} = J\Delta^{1/2}$. The central result of Tomita-Takesaki theory is concerned with this decomposition:

Theorem (Tomita-Takesaki). *Let $\mathcal{N}$ be a von Neumann algebra with Hilbert space $\mathcal{H}$ and cyclic, separating vector Ω . Then, with J and Δ defined as above,*

$$J\Omega = \Omega = \Delta\Omega,$$

and in addition the following equalities also hold

$$\begin{aligned} J\mathcal{N}J^* &= \mathcal{N}' \\ \Delta^{i\tau}\mathcal{N}\Delta^{-i\tau} &= \mathcal{N} \end{aligned}$$

for all $\tau \in \mathbb{R}$, where $\mathcal{N}'$ is the commutant¹⁷ of $\mathcal{N}$ in $\mathcal{B}(\mathcal{H})$ and $\Delta^{i\tau}$ and $\Delta^{-i\tau}$ are defined via functional calculus.

This theorem implies that the $\Delta^{i\tau}$ generate a one-parameter group of automorphisms $\{\sigma_\tau\}$ of $\mathcal{N}$ defined by

$$\sigma_\tau(N) := \Delta^{i\tau}N\Delta^{-i\tau},$$

where N is an element of $\mathcal{N}$. This is known as the modular automorphism group of $\mathcal{N}$ relative to Ω . As we will see, it plays a central role in many recent results in algebraic quantum field theory.

¹⁵ Ω is cyclic if $\mathcal{N}\Omega$ is dense in $\mathcal{H}$.

¹⁶ Ω is separating if for N in $\mathcal{N}$ the equation $N\Omega = 0$ implies $N = 0$.

¹⁷The commutant $\mathcal{T}'$ of $\mathcal{T} \subseteq \mathcal{B}(\mathcal{H})$ is the set of all elements in $\mathcal{B}(\mathcal{H})$ that commute with all elements of $\mathcal{T}$; recall $\mathcal{B}(\mathcal{H})$ is the set of all bounded operators on $\mathcal{H}$.

2.3.4.2 Successes of Tomita–Takesaki Theory The renaissance of Tomita–Takesaki birthed a profusion of beautiful results, far too many to even review here. But, to give a glimpse of their beauty and breadth, we’ll touch upon some highlights. First let’s consider the “KMS condition”.

In quantum statistical mechanics one can define a state ω_ρ using a density matrix ρ and taking traces. In particular, one can consider the state ω_{ρ_β} defined by the density matrix $\rho_\beta = Z^{-1} \exp(-\beta H)$ of the canonical ensemble with Hamiltonian H and inverse temperature β . After a bit of algebra [63], one finds that for any two bounded operators A and B on the Hilbert space

$$\omega_{\rho_\beta}(\alpha_t(A)B) = \omega_{\rho_\beta}(B\alpha_{t+i\beta}(A)),$$

where

$$\alpha_z(A) := e^{izH} A e^{-izH}$$

is the evolution of A under “complex time” in the Heisenberg picture. This relation on ω_{ρ_β} is known as the KMS condition in quantum statistical mechanics, and it survives the thermodynamic limit¹⁸ [63]. Thus it characterizes thermal states [63].

On the other hand, in algebraic quantum field theory if one has a state ω that provides a faithful representation of a unital von Neumann algebra $\mathcal{N}$ via the GNS-construction, i.e. a faithful state, then one can apply the Tomita–Takesaki theorem to the von Neumann algebra $\mathcal{N}$, the GNS-constructed Hilbert space $\mathcal{H}_\omega$ of ω , and the cyclic, separating vector Ω_ω corresponding to the identity operator in $\mathcal{N}$ via the GNS-construction. Doing so one finds [63] that for any A and B in $\mathcal{N}$

$$\omega(\sigma_\tau(A)B) = \omega(B\sigma_{\tau-i}(A)),$$

which exactly mirrors the KMS condition in quantum statistical mechanics if $\beta = -1$. This condition is known as the KMS-condition in algebraic quantum field theory.

This suggests [63] that in algebraic quantum field theory an equilibrium state with inverse temperature β may be characterized as a faithful state whose modular automorphism group $\{\sigma_\tau\}$ is isomorphism to the group of time translations. It’s safe to say, no one saw this coming.

Another success of Tomita–Takesaki was the proof of the CPT theorem [19] within the context of algebraic quantum field theory. Before Tomita–Takesaki theory, the CPT theorem, though proved within the context of Wightman’s axioms, was only conjecture within the context of algebraic quantum field theory. The proof, far too long and detailed to even sketch here, makes extensive use of Tomita–Takesaki theory. In particular, J and Δ are used to implement the CPT operation¹⁹, and properties of J and Δ are then used to show that such a CPT operation is a symmetry.

More recently Tomita–Takesaki theory has led to a proof [12] of the quantum null-energy condition, a lower bound on $\langle T_{ab} k^a k^b \rangle$, the expectation of the energy-momentum tensor contracted against a pair of null vectors. $\langle T_{ab} k^a k^b \rangle$ is proved to be bounded below by a multiple of the second derivative of the von Neumann entropy²⁰. In proving the quantum null-energy condition, Tomita–Takesaki modular automorphisms are used for time evolution, as we saw above for the KMS condition, a critical element in the proof²¹.

¹⁸The thermodynamic limit in this case is the infinite volume, infinite energy, and finite energy density limit.

¹⁹As we have seen, Δ can implement time translations. So J and Δ being able to implement CPT shouldn’t come as a surprise.

²⁰The full details of the bound’s definition are found here [20].

²¹Even more recently Ceyhan and Faulkner published a new proof [30]. Quickly on its heels Hollands and Longo published a follow-up [73] that simplified Ceyhan and Faulkner’s arguments. Both proofs utilize Tomita–Takesaki theory.

2.3.5 Research Questions

Algebraic quantum field theory is vast. Research threads vary from Tomita–Takesaki theory, to holography, to quantum field theory on curved spacetime. There are a vast number of open research questions to pursue. However, the research questions we will focus on will be those related to formalizing algebraic quantum field theory with deep learning in Lean.

So, broadly speaking, our core research question is this: *Can deep learning based interactive proof assistants (IPA) be employed to help formalize algebraic quantum field theory?*

That said, in this section we will only give a “bird’s eye view” of this research question. In later sections we will dive into the details of the questions sketched here.

2.3.5.1 Solidifying Foundations One of algebraic quantum field theory’s key aims is to rise to the challenge of Hilbert’s sixth problem [72], axiomization; so, foundations are critical.

Unfortunately, there are many different variants of “the” Haag-Kastler axioms, e.g. the original Haag-Kastler axioms [64], Haag’s revised axioms [63], Fredenhagen’s version of the axioms [55], Borchers’ version [18], and on and on. It’s unclear which variants are equivalent or inequivalent. It’s also unclear what the variants’ entailment relations are. Broadly put, this leads to the research questions:

- Which Haag-Kastler variants are equivalent?
- What are the entailment relations of the various Haag-Kastler variants?

These research questions need to be answered if the foundations are to be solidified and Hilbert’s sixth problem addressed.

Deep learning based interactive proof assistants (IPA) can assist in answering these questions. For example, if a deep learning IPA is at the level of a researcher, it could be called upon to simply prove the equivalence of two Haag-Kastler variants. Near-term, however, a more likely scenario is that an IPA could be called upon to translate a detailed sketch of an equivalence proof into a complete proof. Another likely, near-term scenario is for an IPA to prove particular lemmas or postulates in a proof sketch.

2.3.5.2 Solidifying Substructure and Superstructure Beyond shoring up foundations, there’s solidifying the substructure and superstructure. Thousands of theorems form the edifice of algebraic quantum field theory. Some of these are derived from one variant of the axioms. Some of them are derived from a completely different variant. Even worse, some of these are derived from a mixture of Haag-Kastler axioms and Wightman axioms²².

The relations between these variants are unclear. The relations between these variants and other axiom systems are also unclear. Thus, it’s completely unclear what theorems are true under what variant(s) or if some theorems are even true under *any* Haag-Kastler variant. This leads to the broad research question: *For any given algebraic quantum field theory theorem, what Haag-Kastler variant does it follow from? (The answer may be: It follows from no variant!)* This obviously needs addressing if algebraic quantum field theory is to aspire to Hilbert’s sixth problem.

As there are thousands of such theorems, this is a *huge* research question, a question far too large for any one person. But, as in the previous section, marshaling deep learning based IPA’s can accelerate progress, hitching progress here to the engine of deep learning, an engine that’s making rapid progress in multiple fields.

²²For example, this is the case for the Bisognano-Wichmann theorem [63] and everything that relies on it!

3 Research Questions

3.1 Theorem Proving with Deep Learning

The overarching research question for theorem proving with deep learning is this: *Can deep learning models match and then surpass human-level, theorem proving abilities?*

Here we discuss (1) benchmarks used to gauge progress on this overarching question, (2) proof generation questions related to this overarching question, and (3) data generation questions related to this overarching question. In a subsequent section we will discuss how deep learning techniques can be employed in formalizing algebraic quantum field theory.

3.1.1 Benchmarks as Bellwethers

While “matching and then surpassing human-level, theorem proving abilities” is the obvious end goal, one must ask: What does this mean? How would we know we’re there? How would we even know we’re going in the right direction? Benchmarks.

Benchmarks are the bellwethers of progress. A curated, peer-reviewed set of benchmarks can function both as a guide, letting us know we are going in the right direction, and as a signpost, letting us know that we have arrived at our destination. With this change of framing, the overarching research question becomes: *Can deep learning models match and then surpass human-level, theorem proving abilities as measured by a curated set of benchmarks?*

While there are numerous benchmarks in the literature—LeanStep [66], LeanDojo [174], MLFMF [15], miniF2F [178], FIMO [95], and ProofNet [11] to name just a few—simply testing progress against all of them isn’t as useful as it may initially seem. Different benchmarks have different strengths and weaknesses. For example, MLFMF [15] extracts data from mathlib [99]; however its main focus is benchmarking premise selection. Also, some are simply out-of-date, targeting Lean 3 instead of Lean 4.

All of this being said, a reasonable selection of benchmarks for this research should include (1) miniF2F for “hard” high school level problems, (2) ProofNet for problems at the level of an undergraduate math major, (3) a benchmark for graduate level problems, and (4) a benchmark for research level problems²³. Unfortunately, graduate level and research level benchmarks don’t exist. However, an upcoming dataset FrontierMath [59] is targeting research level problems, but has yet to be released.

So, in summary, progress on the overarching question will be measured with miniF2F²⁴ and ProofNet²⁵ using various pass@K statistics while relying on the research community to produce in parallel benchmarks for graduate level and research level problems.

3.1.2 Research Questions

Under the umbrella of the overarching research question and in such a new field, the set of questions that can be posed is endless. That being said, we will focus our questions on two topics: proof generation and dataset generation. We’ll further tighten our focus to questions that are answerable in the near-term and that have the highest possible impact.

3.1.2.1 Proof Generation As we’ve previously shown, MCTS algorithms have a high likelihood for impactful, near-term success. So the proof generation research question is: *How can MCTS variants help in answering the overarching research question?*

²³Care must be taken when using miniF2F and ProofNet. Different versions exist.

²⁴Harmonic’s version is reliable, but has non-standard splits. Using it with standard splits is the logical choice.

²⁵Auguste Poiroux’s version is the gold standard.

As you will recall, AlphaGo Zero’s MCTS variant when pitted against its raw policy is expected to loose or draw about 1 out of every 211350 games [128], a strong signal that MCTS can significantly enhance policy performance. In contrast, DeepSeek-Prover-V1.5’s MCTS variant was able to prove 63.5% of the miniF2F test problems as compared to 60.2% proved by the raw policy. This suggests that DeepSeek-Prover-V1.5 is under/improperly utilizing MCTS *or* that MCTS isn’t a good match for theorem proving. However, MCTS has already reached superhuman levels in a wide variety of tasks, e.g. Go, Chess, and Shogi [129]. So under/improper utilization seems more likely, and this suggests that one can hope for significant improvements, spanning orders of magnitude, if MCTS is better utilized.

With this as motivation, our proof generation research questions focus solely on MCTS as it’s highly likely that drastic improvements can be made in the near term. Furthermore, as DeepSeek has publicly released their models along with code for their MCTS variant [171, 119], all of our more fine-grained research questions focus solely on utilizing their framework, as it’ll greatly accelerate progress.

Critic Based MCTS Model The first concrete question we will pose in relation to RMaxTS, the DeepSeek-Prover-V1.5 MCTS variant, is this:

- *Does the introduction of a critic into RMaxTS improve performance on miniF2F and ProofNet?*

In contrast to most modern MCTS variants [128, 129, 85], RMaxTS doesn’t use a critic. This is doubtlessly motivated by GRPO [126], but there is little research to suggest a critic isn’t of use for MCTS variants.

So, motivated by the critics in HPTS [85] and DT-Solver [153], we can sharpen the research question even further:

- *Does the introduction of a “HPTS/DT-Solver type” critic into RMaxTS improve performance on miniF2F and ProofNet?*

Alternative Bandit Algorithms At the core of any MCTS variant sits a particular bandit algorithm, an algorithm selected from the large class of so called bandit algorithms [86]. RMaxTS, the DeepSeek-Prover-V1.5 MCTS variant, is no different in this regard. It is based off of a particular bandit algorithm DUCB [171].

The bandit algorithm selected for a particular MCTS variant greatly influences performance. However, there are hundreds of different bandit algorithms [86] (e.g. UCB, Asymptotically Optimal UCB, MOSS, Bernoulli UCB, Exp3, Exp3-IX...). So the research question arises: *Which bandit algorithm is optimal for RMaxTS?* While optimality is likely too much to ask for, a more focused question is:

- *Which bandit algorithms improve performance of RMaxTS on miniF2F and ProofNet?*

which leads to the follow-up

- *Which bandit algorithms improve performance of a critic based RMaxTS on miniF2F and ProofNet?*

3.1.2.2 Training Data Expansion As training data is the Achilles heel of deep learning, expanding training data is critical. However, human generated datasets are far too small and take far too long to create, and all existing data sets are also unfortunately relatively small. So, we’re forced to generate data. The question is: How?

DeepSeek-Prover-V2 Based MCTS Model Developers of the recently released open source DeepSeek-Prover-V2 models [119] took great care in generating training data.

Using DeepSeek-V3 they decompose a difficult theorem into a set of propositions such that proving all the propositions proved the theorem. They then use a smaller model to prove these propositions. Finally, they assemble these results into a full proof of the original theorem. This process was repeated with a set of difficult theorems to assemble a sizable training data set.

They then used a combination of supervised training and reinforcement learning to train DeepSeek-Prover-V2-671B, a 671 billion parameter model, on this data set. Then they distilled DeepSeek-Prover-V2-671B into a 7 billion parameter model DeepSeek-Prover-V2-7B, a “drop-in” replacement for the core DeepSeek-Prover-V1.5 model [119]. This process guarantees that both DeepSeek-Prover-V2-671B and DeepSeek-Prover-V2-7B implicitly contain knowledge of this sizable training data set. This is evidenced by their SotA results on miniF2F [119].

The obvious research question this poses is:

- *How does RMaxTS perform using DeepSeek-Prover-V2-7B on miniF2F and ProofNet?*

Combining this with our previous questions yields the obvious follow-ups:

- *How does a critic based RMaxTS perform using DeepSeek-Prover-V2-7B on miniF2F and ProofNet?*
- *How does a critic based RMaxTS with an alternative bandit algorithm perform using DeepSeek-Prover-V2-7B on miniF2F and ProofNet?*

Extended MCTS Training Time At the core of most MCTS variants is expert iteration [4]. For a MCTS variant, expert iteration can broadly be described as follows: While training, a MCTS variant may prove, in a previously unseen manner, a set of theorems from the training dataset. These novel proofs are then added to the training dataset, and the policy is trained on this augmented dataset. Hence, the longer training occurs the better the policy becomes as it is trained on more data. The research question this raises is the following:

- *How does performance on miniF2F and ProofNet improve over time for RMaxTS using the core DeepSeek-Prover-V1.5 model?*

This leads to the follow-ups

- *How does performance on miniF2F and ProofNet improve over time for a critic based RMaxTS using the core DeepSeek-Prover-V1.5 model?*
- *How does performance on miniF2F and ProofNet improve over time for a critic based RMaxTS and an alternative bandit algorithm using the core DeepSeek-Prover-V1.5 model?*

Also, these same questions may be posed with the DeepSeek-Prover-V2-7B model.

3.2 Formalizing Algebraic Quantum Field Theory

Complete formalization of algebraic quantum field theory is a massive task, a task which could take any number of directions and give rise to any number of open research questions. So, narrowing scope is critical. Thus, we’ll focus solely upon formalizing the axioms and “foundational” results in algebraic quantum field theory using deep learning and Lean.

With this focus, our overarching research question is this: *Can deep learning based interactive proof assistants (IPA) be employed to help formalize the axioms and “foundational” results in algebraic quantum field theory?*

In particular, here we discuss (1) if formalizing using an IPA makes sense, (2) the IPA based formalization of the various Haag-Kastler variants, (3) and the IPA based formalization of “foundational” results in algebraic quantum field theory.

3.2.1 Does IPA Based Formalization Make Sense?

Does IPA based formalization make sense? We'll address this question by first considering if formalization makes sense, then by considering if IPA based formalization makes sense.

Does formalization make sense? In contrast to many fields within physics, algebraic quantum field theory was conceived as a “pure” mathematical theory *and* as a physical theory. From the beginning it was built upon axioms; these axioms described the physical world and abstract, mathematical concepts. Most other fields within physics are only axiomized decades to centuries after they are discovered, e.g. the symplectic formulation of classical mechanics [6]. So, formalizing algebraic quantum field theory seems to make sense.

However, algebraic quantum field theory, unlike classical mechanics say, is a theory under active development. So “the” axioms change over time. The initial set of axioms, the Haag-Kastler axioms [64], is sometimes used in research while sometimes one of the many variants is used, e.g. Haag’s variant [63] published some 27 years after the initial set. The relations between these variants are unclear. What are the equivalence relations between the variants? What are the entailment relations among the variants? All of these are unanswered. Formalization can bring clarity here, crisply formulating variants and clarifying their relations of equivalence and entailment. So, formalization not only makes sense, it can help to clarify the “core” of the theory.

Does IPA based formalization make sense? A quick look at the scale of the problem will convince one the answer is “yes”. For variants V_i and V_j with $i \neq j$ we can try and prove the entailment relation $V_i \rightarrow V_j$. However, we have to look at all possible entailment relations. The obvious relation $V_k \rightarrow V_k$ along with the variant count, which is about 5 to 10, implies that one has to consider 20 to 90 different entailment relations. Furthermore, if V_j contains $|V_j|$ axioms, then a particular entailment relation $V_i \rightarrow V_j$ can involve up to $|V_j|$ proofs, i.e. assuming each axiom of V_j is a theorem and proving these theorems using the axioms of V_i . The axiom count for the variants runs from about 5 to 10. So the number of proofs required to examine all entailment relations varies from a minimum²⁶ of 20 to a maximum of 900. Without the assistance of an IPA, this is daunting²⁷.

3.2.2 Research Questions

As IPA based formalization seems to make sense, we can narrow our focus. In particular, we'll hone our IPA based formalization questions in on two topics: (1) IPA based formalization of the Haag-Kastler variants and (2) IPA based formalization of “foundational” results. Within these two topics, we will further restrict ourselves to questions with the highest impact that are answerable in the near-term.

3.2.2.1 Haag-Kastler Variants IPA based formalization of the Haag-Kastler variants is a two step process. The first step is manually and/or autoformalizing the axioms of the Haag-Kastler variants. The second step is using our deep learning based IPA to determine their entailment and equivalence relations. As a result we have two initial research questions:

- *Is it possible to manually and/or autoformalize the Haag-Kastler variants in Lean?*
- *Is it possible to utilize our deep learning based IPA to determine variant entailment and equivalence relations?*

As to manual and/or autoformalization, this has the potential to be a labor intensive process. Each of the 5 to 10 variants selected will be formalized by formalizing their 5 to 10 axioms. (Note, we'll

²⁶The value 20 arises in the case of 5 variants and 20 entailment relations. In this case if the first proof attempted for each of the 20 entailment relations fails, one will know each entailment fails and one need not examine subsequent proofs.

²⁷One saving grace is that if $V_i \rightarrow V_j$ and $V_j \rightarrow V_i$, then V_i and V_j are equivalent. So we don't have to consider a separate set of equivalence proofs.

formalize at least the 5 most commonly used variants [64, 63, 5, 18, 55].) In addition, formalization will likely involve introducing new definitions and theorems into `mathlib` and/or `physicslib`—the new open source, GitHub hosted Lean library we will introduce for our work²⁸.

The process of manual and/or autoformalization will involve more than simply transcription. It will also involve clarification. As a concrete example, the original Haag–Kastler axioms [64] include the axiom:

Existence of a Quasi-Local Algebra - The set theoretic union of all $\mathcal{C}(\mathcal{O}_i)$ over $\{\mathcal{O}_i\}$ is a normed $*$ -algebra and its completion $\mathcal{C}$, called the quasi-local algebra, is a C^* -algebra.

where $\{\mathcal{O}_i\}$ is the set of all open sets of Minkowski spacetime with compact closure. As stated, this axiom is under-specified. Is $\mathcal{C}$ the C^* -inductive limit [141] of the $\{\mathcal{C}(\mathcal{O}_i)\}$ or something else? Another example is the original Haag–Kastler axiom:

Local Causality - If $\mathcal{O}_1$ and $\mathcal{O}_2$ in $\{\mathcal{O}_i\}$ are spacelike separated, then $\mathcal{C}(\mathcal{O}_1)$ and $\mathcal{C}(\mathcal{O}_2)$ commute.

This is also under-specified. In what algebra do the $\mathcal{C}(\mathcal{O}_1)$ and $\mathcal{C}(\mathcal{O}_2)$ commute? Is it in the quasi-local algebra $\mathcal{C}$? Is it in some algebra $\mathcal{C}(\mathcal{O}_3)$ where $\mathcal{O}_1 \subset \mathcal{O}_3$ and $\mathcal{O}_2 \subset \mathcal{O}_3$? Similar clarifications will doubtlessly arise in other Haag–Kastler variants.

As to utilizing our deep learning based IPA to determine variant entailment and equivalence relations, this will be one of the first opportunities for our IPA to showcase its abilities. Determining entailment and equivalence relations involves 20 to 900 proofs as we previously determined. So automation will be key. However, each of these proofs will likely not be too “difficult”; so they can provide a testing ground for the initial undergraduate level our IPA will operate at. From these learnings we can further improve our IPA, e.g. using expert iteration [4], to tackle the more “difficult” proofs required when looking into “foundational” results in algebraic quantum field theory.

3.2.2.2 Foundational Results The obvious overarching question when considering our IPA based work on “foundational” results is the following:

Can our deep learning based IPA be employed to help formalize the “foundational” results in algebraic quantum field theory?

This question then leads to the further question: What is a “foundational” result in algebraic quantum field theory?

Generally, foundational results will refer to definitions, theorems, propositions, and lemmas that are covered in, but not limited to, the following references:

- Chapter III of Haag’s *Local Quantum Physics: Fields, Particles, Algebras* [63]
- Chapter IV of Araki’s *Mathematical Theory of Quantum Fields* [5].

We will not attempt to cover all results in these chapters. However, the core definitions will be covered, e.g. state, vacuum, cyclic, separating. In addition, core theorems will also be covered, e.g. GNS construction and the Reeh-Schlieder theorem. Each definition or theorem will be manually and/or autoformalized; again, clarifications will doubtlessly be required. (Time permitting we will also cover more advanced material, e.g. the Tomita-Takesaki theorem [132], DHR analysis [63], and BF analysis [63].)

Once formalized an attempt to answer the overarching question will be made by employing our IPA to prove the theorems. Proofs will occur in a four step process. First, the IPA will simply be given the

²⁸This library is an alternative to the `PhysLean` library [148] which covers physics topics, but doesn’t take full advantage of Lean’s strengths.

opportunity to generate a proof for the theorem. Second, for those theorems it was unable to prove, a formalized proof blueprint will be manually and/or automatically created. Third, the IPA will then be given the opportunity to generate a proof for all propositions in the proof blueprint. Fourth, for those propositions it is unable to prove, it will be given the opportunity have a human guide its proof step generation towards a proof of each such proposition. In addition each proof will be attempted using each Haag–Kastler variant equivalence class, firmly establishing which theorems follow from which Haag–Kastler variants.

4 Research Objectives

4.1 Theorem Proving with Deep Learning

Open source software built around an active community can change the world. One need look no further than AlphaFold [46] which revolutionized drug discovery [28] and led to the 2024 Nobel Prize in Chemistry [146].

Motivated by this example, our global objective is to open source a Lean integration of our SotA deep learning model and to build an active community around the integration. Other objectives flow naturally from this: (1) achieving SotA benchmarks, (2) integrating our model into Lean, and (3) building a community around the integration. While these objectives are ambitious, they are also firmly grounded.

4.1.1 Research Objectives

MCTS, as we have seen, is likely underutilized in theorem proving. So, pushing the SotA envelope with MCTS seems achievable. Furthermore, in building upon the open source DeepSeek-Prover-V1.5 framework [171] we will be able to significantly accelerate development timelines.

As to Lean integration, Lean CoPilot [131] open sourced a framework for integrating deep learning models into Lean. It allows for both proof step generation and proof generation. In addition Panto-graph [3] also provides a framework for integrating deep learning models. So, there are multiple paths to our objective.

In regard to building a community, Lean already has a strong community, as evidenced by the activity on their Zulip [89] channels. Leveraging such a community is something the author has much experience with. Heading up the machine learning group at Mozilla, one of the world’s largest open source companies, and managing open source projects that have garnered over 80k stars on GitHub [102, 103, 38, 37, 36] provided an abundance of community building experience.

4.1.1.1 miniF2F Benchmark

Objective The miniF2F benchmark contains olympiad-level problem statements from AMC, AIME, and IMO²⁹ as well as other high school and undergraduate level problems [179]. So mastering this benchmark will indicate abilities on par with a gifted high school student.

On the leaderboard Kimina-Prover-Preview [154] holds the current SotA result [113] for the miniF2F test set, solving 80.74% of the problems at pass@8192. DeepSeek-Prover-V2-671B, though not on the leaderboard yet, solved 88.9% and DeepSeek-Prover-V2-7B solved 82.0% both at pass@8192. So the DeepSeek-Prover-V2 models are SotA.

²⁹AMC, AIME, and IMO are mathematics competitions for gifted high school level competitors.

At a minimum, our objective is to surpass the 88.9% result on miniF2F test. However, our “stretch” objective is 100.0%. As these problems are at the level of a gifted high school student and our ultimate goal is a superhuman result, we should strive to solve all of the problems.

Importance The miniF2F benchmark is commonly used and thus a good indicator of relative (and absolute) progress. Hence, surpassing SotA is a notable signpost of success. In addition, the skills the model learns in achieving this objective will come in good stead proving results related to algebraic quantum field theory. However, these skills are at the level of a gifted high school student, as opposed to say an undergraduate math major; so they won’t be all of what one needs for work on algebraic quantum field theory.

4.1.1.2 ProofNet Benchmark

Objective ProofNet is a benchmark that contains problems from undergraduate, pure mathematics textbooks [11]. It covers real and complex analysis, linear algebra, abstract algebra, and topology among other topics [11]. Hence, this benchmark indicates abilities on par with an undergraduate math major.

DeepSeek-Prover-V2-671B [119] appears³⁰ to have the SotA result on the ProofNet test set, solving 37.1% of the problems at pass@1024. Again, DeepSeek-Prover-V2-7B [119] isn’t far behind with 29.6% at pass@1024. So, yet again, the DeepSeek-Prover-V2 models are SotA.

Minimally our objective here is to surpass the 37.1% SotA result. Again, our “stretch” objective is 100.0%. As these are undergraduate level problems and our ultimate target is superhuman performance, 100.0% should be our ultimate objective.

Importance As with miniF2F, ProofNet is a common benchmark and thus a solid measure of our progress in relative (and absolute) terms. Surpassing SotA lets us know we’re on the right track, but doing so isn’t our ultimate goal. These problems are “pure” mathematics problems and thus more similar to what’s encountered in algebraic quantum field theory than say those of miniF2F. However, these problems are “only” at the level of an undergraduate math major. Hence, they are not our final destination.

4.1.1.3 Lean Integration

Objective Lean CoPilot [131] and Pantograph [3] allow for a deep learning model to be integrated into Lean.

Lean CoPilot allows for proof step generation and proof generation [130] with the model being either local or in the cloud [130]. Lean CoPilot exposes the model through the commands `suggest_tactic` that queries the model for the next proof step and `search_proof` that queries the model to generate an entire proof.

Pantograph, on the other hand, allows for MCTS and a draft-sketch-proof (DSP) approach [82] to theorem proving. Draft-Sketch-Proof (DSP) allows a model to create a proof blueprint consisting of a sequence of have statements, each concluded with a sorry placeholder, then to prove each of these statements one by one. This allows a model to decompose a complex problem into a sequence of simpler problems, much the way mathematicians work.

Our objective is to integrate our deep learning model into Lean using Pantograph [91] or Lean Copilot [130]. This integration will allow the model to run locally, downloading the model from say

³⁰There appears to be no ProofNet leaderboard. So, the result here comes from looking at the results of all articles that reference the ProofNet paper.

Hugging Face, or in the cloud, using say a Hugging Face endpoint. We will also package the code so it's available to end users via an `import` statement after they add the package as a dependency.

Importance Integration into Lean is critical for adoption. Users have no patience for setting up a Python environment, downloading an external model, installing that model, extracting their code out of Lean, importing that code into a model... Users will not spend an inordinate amount of time “yak shaving”³¹ to test our model. They want immediate results.

So, providing a package that is effortlessly installed and is easy to use will increase adoption multifold. Furthermore, the simple, minimal interface of, say, a `suggest_tactic` and `search_proof` will also buoy adoption. All of these “quality of life issues” will be critical to the long term success of the project.

4.1.1.4 Community

Objective Deep learning is advancing at breakneck speed. No one person, or organization for that matter, is able to participate in, or even stay abreast of, all advances in the field. Only a community of individuals and organizations has any hope of keeping up. This is not only the case for deep learning, but also the case for deep learning theorem proving.

So, our objective is to build a large, vibrant community around our open model, open Lean integration, and open formalization work. It is only in doing so that the project has hope of staying at the cutting edge and thus being of maximal use to the physics and mathematical communities.

Importance Long term the benefits afforded by deep learning and by extension deep learning theorem proving will make it all but impossible for mathematical physicists to ignore them. They will integrate such tools into their workflow, or risk being left behind. A similar revolution has already occurred in computational biology with the introduction of AlphaFold [46], and analogous transformations are inevitably coming to other fields too.

Catalyzing this revolution in mathematical physics, as we intend to do, is critical. But, as no one person or organization can stay on top of all advances in the field, building a community to guide this overarching transformation is of utmost importance. A community will stay abreast of new developments; help to integrate newer, more performant models; identify and fix bugs; help with documentation; suggest improvements. . . The list is endless and critical to the success of the overarching goal.

4.1.2 Relevance, Importance, and Impact

These objectives are not only salient to this project, they are also salient to the global goals of this research thread.

4.1.2.1 Importance The arc of progress in deep learning is such that it's not a question of “if”, but “when” deep learning tools become indispensable to mathematical physicists [144]. The building blocks for these future tools are here now (e.g. Lean's `mathlib` [99], DeepMind's AlphaGo [129], and DeepSeek's DeepSeek-Prover-V1.5 [171] and DeepSeek-Prover-V2 [119]), and this inevitable future demands they be assembled. Our project assembles these tools into a nascent form of what could be, then builds a community to shepherd this nascent effort towards its inevitable future. Such work is indispensable.

³¹Yak shaving is engaging in a long series of tangentially, unrelated, or unnecessary tasks before tackling the main task.

4.1.2.2 Relevance Our objectives, while future looking, hold concrete relevance to many current threads of research. For example, our objective of progress on the miniF2F and ProofNet benchmarks is critical to the larger research community’s progress. Similarly, the objective of Lean integration is relevant for driving adoption of deep learning tools in Lean; easy to install and use it widens acceptance and broadens the user base. Finally, if the mathlib community is anything to go by, the acceleration afforded by leveraging community is vital for progress in large, long-term objectives. So, the objective of building community is relevant to accelerating progress of the project and the field as a whole.

4.1.2.3 Impact All strategies and tactics involved are chosen to maximize impact. Open sourcing the models, the Lean integration, and the Lean library ensure frictionless dissemination (e.g. downloading models from Hugging Face) and extensive exploitation (e.g. building on our code in GitHub), essential for maximizing impact. Community too maximizes impact. GitHub is laden with single developer repositories that have no community and no impact. Community spreads the word, assisting in dissemination and thus impact. Community gives back too, adding enhancements that improve the code base. This further enhances impact and widens the community, creating a virtuous cycle.

4.2 Formalizing Algebraic Quantum Field Theory

If we are to rise the challenge of Hilbert’s sixth problem [72],

...diejenigen physikalischen Disziplinen axiomatisch zu behandeln, in denen schon heute die Mathematik eine hervorragende Rolle spielt ³²,

a sound foundation is indispensable. However, with the proliferation of Haag-Kastler variants, we have anything but.

Galvanized by Hilbert’s call to action, our overarching objective is to open source a Lean library formalizing algebraic quantum field theory (Haag–Kastler axioms, foundational results, and (maybe) advanced results) and to build an active community around this library. Subsidiary objectives logically follow: (1) formalizing Haag–Kastler variants, (2) determining variant entailments, (3) determining variant equivalences, (4) formalizing foundational results, (5) (maybe) formalizing advanced results, (6) creating and open sourcing a Lean library containing these results, and (7) building a community around this library. While these are ambitious objectives, when accelerated by our deep learning IPA, we can turn these ambitions into reality.

4.2.1 Research Objectives

Haag-Kastler variants [64, 63, 5, 18, 55] sew confusion. Clarity of *the* Haag-Kastler axioms [64] is lost. Our objective is to revive this clarity. However, such clarity can only return if variants are formalized and entailments and equivalences proven. This objective, while ambitious, is achievable if facilitated by our deep learning IPA.

The confusion the variants sew extends to foundational (and advanced) results. Which theorems rely on which variants? Our objective is to eliminate this confusion. Doing so requires formalizing foundational (and advanced) results then determining which Haag-Kastler variant equivalence classes³³ they follow from. Aided by our deep learning IPA, this ambitious objective emerges as achievable.

Motivated by mathlib library’s [99] success, an additional objective is to package all appropriate definitions, axioms, theorems, and proofs into a new, open source library physicslib. In contrast to the

³²Translated: “...to treat those physical disciplines in which mathematics already plays a prominent role axiomatically”

³³The equivalence class $[V_i]$ consists of all variants V_j such that $V_i \leftrightarrow V_j$. Thus, if V_i and V_j are in the same equivalence class, then we need only determine if a theorem follows from say V_j as $V_i \leftrightarrow V_j$ implies the theorem also follows from V_i .

nascent, physics library PhysLean [148], our library physicslib will only contain physics theories that rise to the challenge of Hilbert’s sixth problem and are axiomized.

Community is key to mathlib; so building a community will be a key physicslib objective. Leveraging mathlib’s community in this objective is crucial; their experience can steer us clear of any hidden dangers they discovered along the way. Also, the author’s experience leading up multiple, major open source projects [102, 103, 38, 37, 36] will also serve in good stead on this journey.

4.2.1.1 Haag–Kastler Variant Formalization

Objective Since the publication of the original Haag-Kastler axioms [64], many variants [64, 63, 5, 18, 55] have arisen. These variants differ in many subtle, but possibly impactful, aspects, and exist in differing levels of rigor.

Our objective is to manually and/or autoformalize at least the 5 most commonly used variants [64, 63, 5, 18, 55] in Lean. Variants’ differing levels of rigor imply formalization will not only involve transcription, but also clarification. Furthermore, formalization will also entail enhancing mathlib with “squarely” mathematical definitions and theorems, e.g. C^* -inductive limit associated with algebraic quantum field theory.

Importance If we are to rise to the challenge of Hilbert’s sixth problem, foundations are critical. Hence, clear, unambiguous declarations of the Haag-Kastler variants are indispensable. Lean formalization provides those declarations. Furthermore, through formalization, the Haag-Kastler variants become computable, opening up lines of research (some of which we can’t foresee) which heretofore were impossible.

4.2.1.2 Haag–Kastler Variant Entailment

Objective With the proliferation of Haag-Kastler variants, each differing in subtle ways, the entailment relations between these variants must be finely-drawn. But as of yet, it’s unclear what these finely-drawn entailment relations actually are.

Our objective is to clarify the precise entailment relations among the Haag-Kastler variants, i.e. to prove all variant entailment relations in Lean. As we’ve seen, this involves an inordinate number of proofs, a minimum of 20 to a maximum of 900. Manually formalizing 900 proofs is daunting. However, with the assistance of our IPA, this task can thankfully be cutdown to a human level.

Importance Without clarity on variant entailment, the foundations (plural), substructures (plural), and superstructures (plural) of algebraic quantum field theories (plural) are uncertain. Establishing entailment relations clarifies this confusion. Entailment relations will allow all results from variant V_j ’s thread of research to immediately be subsumed into variant V_i ’s thread by simply establishing that V_j is entailed by V_i . Clarity.

4.2.1.3 Haag–Kastler Variant Equivalence

Objective Minute variant variations breed uncertainty regarding equivalences. Does using a net of C^* -algebras based on bounded, open sets differ from using one based on casually complete sets? As of yet, no clarity exists on the subtle effect these minute differences have on variant equivalences.

Our objective, thus, is to illuminate the exact equivalence relations among the Haag-Kastler variants, i.e. to prove all possible variant equivalence relations in Lean. Luckily this work can piggyback on

entailment; $V_i \rightarrow V_j$ and $V_j \rightarrow V_i$ imply equivalence. Hence, the inordinate number of entailment proofs serve double duty.

Importance Establishing equivalence relations will fuse seemingly disparate threads of research. Results of older variants can follow from newer. Results of newer variants can follow from older. Proofs, complicated under the light of one variant, may appear simple when viewed under the light of an equivalent variant. Broadly, establishing all equivalences will cut through much of variants’ complexity, simplifying foundations, substructure, and superstructure of algebraic quantum field theory.

4.2.1.4 Foundational Results

Objective Foundational results³⁴, while largely similar across variants’ research threads (e.g. some version of the Reeh-Schlieder theorem always makes an appearance), tend to differ in subtle and sometimes not so subtle ways across variants. For example, the definitions of “state”, “vacuum state”, “cyclic”, and “separating” are largely variant independent. However, the Reeh-Schlieder theorem can differ significantly across variants. For instance, in Haag [63] its proof depends upon the Wightman axioms while in Araki [5] its proof depends only upon a Haag-Kastler variant. These are *material* differences.

Our objective then is to formalize these foundational results in Lean. Subsidiary objectives naturally follow: (1) identifying and manually and/or autoformalizing foundational definitions; (2) identifying and manually and/or autoformalizing foundational theorems, propositions, and lemmas; and (3) utilizing our IPA to formalize proofs of these foundational propositions over all possible variant equivalence classes. As illustrated by the Reeh-Schlieder theorem, identifying what to formalize isn’t simply an act of transcription. It involves efforts beyond the act of rote formalization.

Importance Foundational results support the superstructure; so they too must be sound, depending only upon a variant’s axioms. However, in the literature this is not always the case, e.g. the Bisognano-Wichmann theorem[63] is “proved” using a mixture of Haag-Kastler axioms and Wightman axioms. Hence, formalizing the statements and proofs of foundational results using only Haag-Kastler variants significantly solidifies the entire superstructure, something the entire field will, and needs to, profit from.

4.2.1.5 Advanced Results (Stretch Objectives)

Objective Algebraic quantum field theory rises far above the foundation and substructure of the Reeh-Schlieder theorem and GNS-construction. The CPT theorem [18], Tomita-Takesaki theorem [132], DHR analysis [63], BF analysis [63], and the Bisognano-Wichmann theorem[63], just to name a few, are vital to algebraic quantum field theory’s superstructure. Doubtlessly, these should be included in any comprehensive treatment.

That being said our (stretch) objective is to formalize some of these advanced results in Lean. Again, subordinate objectives immediately follow: (1) identifying and manually and/or autoformalizing required definitions; (2) identifying and manually and/or autoformalizing advanced theorems, propositions, and lemmas; and (3) utilizing our IPA to formalize proofs of these advanced propositions over all possible variant equivalence classes. As in the case of the foundational results, this involves far more than rote formalization.

³⁴For our purposes “foundational results” will refer to definitions, theorems, propositions, and lemmas that are covered in, but not limited to, Chapter III of Haag [63] and Chapter IV of Araki [5].

Importance As is the case for the foundation and substructure, formalizing these advanced results will shore up the superstructure of algebraic quantum field theory, benefitting the field as a whole. Results upon which many have built, e.g. Bisognano-Wichmann theorem [63], will (hopefully) be established fully within the auspices of algebraic quantum field theory, removing any doubts about those results and, by extension, about algebraic quantum field theory as a whole.

4.2.1.6 Lean Library `physicslib`

Objective Formalization efforts in the `mathlib` library [99] have gone from strength to strength. Covering virtually all the territory of undergraduate mathematics, significant swaths of graduate studies, and islands of research mathematics, its scope is testament to its success.

Inspired by this success, our objective is to introduce `physicslib`, a Lean library focused exclusively on physics or, more to the point, on physics that rises to Hilbert’s challenge of axiomatization. Our axiomatization of algebraic quantum field theory will be the inaugural addition. However, with community help, numerous other axiomizations could be added, e.g. Wightman’s axioms [133], Atiyah–Segal axioms [7], Turaev–Blanchet axioms [16], or any other number of axiomized physical theories.

Importance The `mathlib` library is a fixed point around which the mathematics community rallies, sharpening definitions, proofs, and propositions. `physicslib` can play that role for physicists. A gathering point for axiomatic physics, a repository of axioms and rigorous results, a fountainhead of deep learning training data, and a building block for future results we can only dream of.

4.2.1.7 Community

Objective Physics is an immense field of study, from stellar interiors to self-organizing criticality. Today, no one person could hope to master its breadth and depth. Federating individuals and organizations is the only hope of embracing its magnitude. This is true not only of physics in general, but it’s true of axiomized physics in particular.

Our objective, thus, is to build a large, diverse, vibrant community around `physicslib`. This is the only way to cover the breadth and depth of axiomized physics and rise to Hilbert’s sixth problem.

Importance Community is essential. Without community, `physicslib` goes the way of the `PhysLean`; with only 10 contributors [70], it’s interesting but lacks impact.

Community serves many objectives for an open source project. For example: (1) through word-of-mouth it aids dissemination; (2) through Linus’s law³⁵ it exposes problems early; (3) through diversity it covers physics’ breadth; (4) through size it probes physics’ depth; (5) through sheer number it becomes self-sustaining; and (6) through its dimension it drives impact. Community is indispensable.

4.2.2 Relevance, Importance, and Impact

These objectives are not only germane to the local arc of this project; they are germane to the global arc of algebraic quantum field theory, quantum field theory, and physics as a whole. Paired with deep learning proof assistants and the ever accelerating pace of deep learning, they have the real possibility of changing the practice of physics forever.

³⁵Linus’s law states: “With enough eyeballs all bugs are shallow”.

4.2.2.1 Importance High energy theoretical physics operates under an increasing load of fundamental principals, e.g. Lorentz invariance, and a decreasing load of experimental results, e.g. physics beyond the Standard Model. Thus, it's evermore critical these fundamental principals be cleanly drawn and the conclusions they entail be sound. Axiomizing physics in Lean accomplishes both.

Fundamental principals are captured in axioms while conclusions' soundness is guaranteed by the Lean runtime. Traditionally such tech has been shunned; the velocity of results it affords is modest. However, with the arc of progress in deep learning and the advent of deep learning based IPA's, this will change. It's not a question of "if" but "when" such tools become indispensable to mathematical physicists [144]. The importance of our work follows then from being the foundation of this inexorable future.

4.2.2.2 Relevance Algebraic quantum field theory is in the midst of a second renaissance [166, 134, 165, 135, 31, 97, 164, 73, 96]. Notably, its fevered pitch has pierced the public's veil, with popularizations reaching even Quanta Magazine [32]. Our work holds relevance to this second renaissance, as well as to the first. Formalization fortifies foundations, bringing certainty to old results and pointing the way towards new ones. Moreover, deep learning will quicken this wave of formalization, widening the reach of its crests and troughs.

4.2.2.3 Impact Impact was top of mind when planning strategies and tactics. Algebraic quantum field theory, in the midst of a second renaissance, is topical; any novel work there attracts attention, maximizing impact. physicslib too maximizes impact. Open sourcing expands dissemination (e.g. frictionless integration) and aids exploitation (e.g. anyone can fork our code) which amplify impact.

Community maximizes impact and impact maximizes community. As is evident, *community maximizes impact*, i.e. our work influences the group of individuals and organizations that constitute our community. Surprisingly, the "inverse" also holds; *impact maximizes community*, i.e. excited about the work our community spreads the word bring others into the fold. This is the virtuous cycle of open source, our engine for impact.

5 Methodology

We divide methodology into six work packages: two theorem proving work packages—Proof Generation (WP1) and Training Data Expansion (WP2)—three formalizing work packages—Variant Formalization (WP3), Foundational Result Formalization (WP4), and Advanced Result Formalization (WP5)—and one dissemination and exploitation work package—Dissemination and Exploitation (WP6). The theorem proving work packages describe the approach we will take to address the research questions raised in subsection 3.1 Theorem Proving with Deep Learning. The formalization work packages describe the approach we'll take in addressing the research questions of subsection 3.2 Formalizing Algebraic Quantum Field Theory. The dissemination and exploitation work package covers our approach to research objectives of paragraph 4.1.1.4 and paragraph 4.2.1.7.

The theorem proving work packages create the deep learning models and Lean integration which accelerates work of the formalizing work packages. Hence, theorem proving work packages kick-off the project. Instead of starting from zero, the theorem proving work packages use as their foundation DeepSeek-Prover-V1.5 [171], a MCTS variant and suite of open source models that perform near SotA on several benchmarks, and DeepSeek-Prover-V2 [119], a pair of open source models that perform at SotA on several benchmarks.

The formalizing work packages start in earnest only after the theorem proving work packages are

complete. This allows for the most performant models and the most advanced Lean integration to be utilized in the formalizing work packages and thus speeds the formalization process.

The dissemination and exploitation work package is woven throughout the entire project. For example, when the project reaches a particular milestone, blog and social media posts will announce the result. Or, for example, we will present at conferences throughout the lifetime of the project to raise its profile and thus broaden community.

5.1 Theorem Proving with Deep Learning

Together the two theorem proving work packages Proof Generation (WP1) and Training Data Expansion (WP2) outline our plan of attack for the research questions raised in subsection 3.1 Theorem Proving with Deep Learning. In particular the Proof Generation (WP1) work package sets out our approach to research questions of paragraph 3.1.2.1 Proof Generation. While the Training Data Expansion (WP2) work package speaks to our plans for the research questions of paragraph 3.1.2.2 Training Data Expansion.

5.1.1 WP1: Proof Generation

Far from lacking foundation, this work package builds on the foundation of DeepSeek-Prover-V1.5 [171] and DeepSeek-Prover-V2 [119]. DeepSeek-Prover-V1.5 [171] consists of a pre-trained model, supervised fine-tuned model, and reinforcement learning model as well as code for RMaxTS, the MCTS variant used in DeepSeek-Prover-V1.5 [171]. DeepSeek-Prover-V2 [119] consists of two models, one a 671 billion parameter model the other a 7 billion parameter model, both at or near SotA.

RMaxTS MCTS consists of four phases selection, expansion, simulation, and backup [40]. MCTS variants take this general framework and alter aspects to better suit a particular problem. RMaxTS, in particular, alters the selection, expansion, and simulation phases in significant ways.

In the selection phase RMaxTS introduces a novel tree selection policy that selects not only leaf nodes, as is standard in MCTS, but internal nodes too, which isn't standard in MCTS. The tree selection policy it uses is based off of DUCB [57], a variant of UCB1 [8], that uses a discount factor to smoothly drop "outdated" rewards. This is necessitated as RMaxTS introduces a non-stationary reward. It is the sum of a stationary reward for finding a proof, which is standard, and a non-stationary reward for adding a new node to the MCTS search tree, which is not standard.

RMaxTS merges the expansion and simulation phases. In MCTS the expansion phase normally adds a new, single child to the node selected in the selection phase. Normally the simulation phase then runs a simulation to assign a reward to this new, single child. RMaxTS, in contrast, adds a new subtree to the node selected in the selection phase. If this subtree completes the proof, then RMaxTS gives the proper reward; if it doesn't complete the proof, it gives no reward. RMaxTS does this all "in one go", merging the MCTS expansion and simulation phases.

The RMaxTS alterations of the selection, expansion, and simulation phases are not without logic. The selection of internal nodes during the selection phase, the addition of subtrees in the expansion phase, and the merging of the expansion and simulation phases are all well motivated.

For example, on selection, MCTS is normally targeted towards games that at any point have a finite number of valid possible next moves. In contrast in a proof there are always an infinite number of valid possible next steps, e.g. introducing a new real valued constant. Hence, the standard MCTS assumptions no longer hold. In particular it's always possible to add a new child to an internal node.

On expansion and simulation, the RMaxTS design is guided by the relatively high cost of inference and calling into the Lean runtime. Standard MCTS would introduce a new, single node. Creating and validating this single node necessitates inference and a call to the Lean runtime which is costly. RMaxTS,

in contrast, with a single inference introduces an entirely new subtree while only necessitating a single call to the Lean runtime, a gain in efficiency. In addition, the call to the Lean runtime indicates if the subtree proves or doesn't prove the theorem. Hence, the reward doesn't require a separate simulation phase.

RMaxTS Problems While the RMaxTS alterations are well motivated, they also introduce problems. These problems, far from damming the approach, present opportunities to improve RMaxTS.

First among the problems is the sparsity of reward. You are only rewarded if the proof is successful. This is sparse. They try to counteract this by introducing a non-stationary reward for adding a new node to the MCTS search tree. However, the core problem remains. Normally MCTS solves this problem by introducing a critic. RMaxTS doesn't use a critic.

Second, RMaxTS's DUCB based tree selection policy is suboptimal. DUCB ignores the prior provided by the deep learning model, in contrast to say PUCT [128], and relies upon an accumulation of node statistics to guide node selection. However, "minimal" node statistics are gathered. The maximum number of times any node is visited³⁶ is 32 [171]. Relative to the infinite number of possible next steps in a proof, 32 is minuscule.

Through the various tasks of this work package, we address the above, and other, shortcomings of RMaxTS, improving its performance on the miniF2F and ProofNet benchmarks.

5.1.1.1 T1.1: Critic Based MCTS Model

Research and Development Sparsity of reward is a critical problem for RMaxTS. Allotting rewards only upon proof completion provides insufficient feedback. Hence, RMaxTS introduces an auxiliary, non-stationary reward for adding new nodes to the MCTS tree. This auxiliary reward helps, but it doesn't provide sufficient feedback. All new nodes are not of equal "value". A new node that is more likely to lead to a proof should receive a higher reward than a new node that leads to a "dead end". This obvious distinction is not made by RMaxTS.

To fix this we propose introducing a critic into RMaxTS, doing away with the auxiliary reward, and replacing DUCB with UCB1³⁷. The critic, when presented with a partial proof tree, assigns a reward to the partial proof tree. The higher the reward the more likely the critic "believes" the partial proof tree can lead to a complete proof tree. This critic is integrated into RMaxTS by separating the expansion and simulation phases. The expansion phase adds a subtree as before. If the proof is incomplete after addition of the subtree, the simulation phase uses the critic to assign a reward to the partial proof tree, root node down to and including the newly added subtree. If the proof is complete, the simulation phase allots reward 1.

One reason DeepSeek-Prover-V1.5 [171] didn't introduce a critic was that doing so "required" an entirely new model be developed and trained. A new model is not required. As in HTTPS [85], the main model can be made to operate in at least two different modes. In the first "standard" mode, presented with a partial proof tree it outputs Lean code in an effort to complete the partial proof tree. In the second "critic" mode, presented with a partial proof tree prefixed by the special token CRITIC it outputs the tokens PROVABLE and UNPROVABLE with probabilities that add to one. The reward it assigns to a partial proof tree is then the probability the model assigns to the output PROVABLE.

³⁶Examine the DeepSeek-Prover-V1.5 rows of the "Tree Search Methods" section of Table 1 in [171]. The sample budget K is presented as $K = N \times T$ where N is the number of tree search attempts. N varies from 1 to 32.

³⁷Originally DUCB was used in place of UCB1 because the reward was non-stationary [171]. As we remove the non-stationary reward, UCB1 can be put back in place.

Training Training will proceed using the LightZero [105] training framework³⁸ in a manner largely described in the original DeepSeek-Prover-V1.5 [171] research. Introduction of the critic, however, complicates this a bit.

As in HTPS [85] the critic will be trained to minimize the cross-entropy loss of the restricted output tokens PROVABLE and UNPROVABLE conditioned on an input partial proof tree. The (*partial proof tree, reward*) pairs used to train the critic will be selected as in HTPS [85] from complete and incomplete proofs created while training. In particular: (1) selecting partial proof trees that are part of a completed proof and assigning reward 1 to them (2) selecting partial proof trees that are not part of a completed proof, but which have visit count greater than a certain threshold, and assigning a reward based on the empirical action value of their final node.

Benchmarking Benchmarking will be against the test sets of miniF2F [179] and ProofNet [11] using pass@k metrics³⁹. As the RMaxTS variant described here utilizes UCB1 which relies upon an accumulation of node statistics, benchmarking will allow the algorithm “sufficient” inference time to accumulate “sufficient” node statistics to guide node selection.

Timetable As this is the first task, extra time must be allocated to allow for setting up the training pipeline, say one month to simply reproduce the DeepSeek-Prover-V1.5 [171] pipeline. In addition the critic must be created and integrated into this pipeline. As the critic itself is relatively simple, the majority of time will be spent integrating it into the pipeline and training, say an additional two months. So the entire time required for T1.1 is 3 months.

5.1.1.2 T1.2: Alternative Bandit Algorithm

Research and Development Improper and/or under utilization of the DUCB tree policy is a critical problem for RMaxTS. Accuracy of the DUCB tree policy for a given node relies upon collecting a statistically significant sample of data for each child. As any MCTS tree node has an infinite set of possible children, collecting such a statistically significant sample is a fool’s errand.

To fix this we propose introducing an alternative tree policy. There are hundreds of possible policies [86]. A smaller set of these are targeted towards bandits with an infinite number of arms, i.e. the case we are dealing with. Of this smaller set, the one with the most success in this arena, and the one we will explore first, is PUCT [128].

Conditioned on a partial proof tree, PUCT introduces a model that provides a prior over the set of possible children of the final node in the partial proof tree. This prior is then used in the PUCT tree policy to select possible children during the selection phase of MCTS. So we will exchange DUCB for PUCT but keep the non-stationary, auxiliary reward along with the discount factor that discounts older rewards.

One might think that a new model is required for the PUCT tree policy. This is not the case. The main model in its “standard” mode of operation if presented with a partial proof tree outputs Lean code in an effort to complete the partial proof tree. If one retains only the initial proof step in the Lean code output, then this initial proof step can be interpreted as selecting a child of the final node in the partial proof tree. So, no extra model is required.

Training As PUCT is simply repurposing the main model, training will proceed as described in original DeepSeek-Prover-V1.5 [171] research but using the LightZero [105] training framework⁴⁰.

³⁸We will also explore use of 3FS [47] to run LightZero across multiple nodes.

³⁹pass@k for $k \in \{1, 32, 64, 100, 1024, 8192\}$ seem to be the most commonly used values [113].

⁴⁰Again, we will explore using 3FS [47] to run LightZero across multiple nodes.

Benchmarking Benchmarking will be against the test sets of miniF2F [179] and ProofNet [11] using pass@k metrics. As UCB1 isn't being used in this case, benchmarking need not make such a concerted effort to allow the algorithm "sufficient" inference time to accumulate statistics.

Timetable As the PUCT model is simply repurposing the main model, the majority of time will be spent integrating it into the pipeline and training. This will require, say, two months. So the entire time required for T1.2 is 2 months.

5.1.1.3 T1.3: Critic + Alternative Bandit Algorithm Sparsity of reward and improper and/or under utilization of the DUCB tree policy are the critical problems for RMaxTS. As we've seen, a critic can help with reward sparsity and the PUCT tree policy can help with improper and/or under utilization of DUCB. So combining a critic and PUCT can help with both problems. This task does just that.

Research and Development As this task combines the results of T1.1 and T1.2, there is no need for research and development. One need only integrate the products of T1.1 and T1.2.

Here DUCB of RMaxTS will be replaced with PUCT and the non-stationary, auxiliary reward will be replaced by the critic's feedback. Furthermore, as the critic's reward is stationary, there's no need for discounting the rewards.

Training As this task is simply repurposing the products of T1.1 and T1.2, training will proceed as described in T1.1, where the pipeline was modified to allow for training a critic.

Benchmarking Benchmarking will be against the test sets of miniF2F [179] and ProofNet [11] using pass@k metrics and proceed as in T1.2.

Timetable As this task is simply repurposing the products of T1.1 and T1.2, the majority of time will be spent integrating the products into the pipeline and training. This will require, say, two months. So the entire time required for T1.3 is 2 months.

5.1.2 WP2: Training Data Expansion

High-quality training data for theorem proving is in short supply. Compounding this problem is the fact that manually creating new training data requires a high-skill level, e.g. mathematics professors, and is a time-consuming process. Automatic data creation, however, cuts this Gordian knot.

Automatic data creation was central to the developers of the DeepSeek-Prover-V2 models [119]. They took care in creating data from difficult theorems through a DSP-like process, ending up with a sizable data set. They then trained two models DeepSeek-Prover-V2-671B, a 671 billion parameter model, and DeepSeek-Prover-V2-7B, a 7 billion parameter model, on this data set. The models DeepSeek-Prover-V2-671B and DeepSeek-Prover-V2-7B gained implicit "knowledge" of this sizable data set.

To expand training data this work package exploits these facts: (1) DeepSeek-Prover-V2-7B is a "drop-in" replacement for the core DeepSeek-Prover-V1.5 model and (2) DeepSeek-Prover-V2-7B has implicit "knowledge" of the sizable data set created by the DeepSeek-Prover-V2 developers. Together these imply that DeepSeek-Prover-V2-7B can be "dropped-in" to the frameworks of WP1 while utilizing its implicit "knowledge" to improve performance over DeepSeek-Prover-V1.5. This work package explores this in detail.

5.1.2.1 T2.1: DeepSeek-Prover-V2-7B Based MCTS Model

Research and Development The initial step of T1.1 was to reproduce the DeepSeek-Prover-V1.5 pipeline. This current task will simply replace the core DeepSeek-Prover-V1.5 model with DeepSeek-Prover-V2-7B in that pipeline. In that pipeline DeepSeek-Prover-V2-7B can leverage RMaxTS to improve its performance relative to the core DeepSeek-Prover-V2-7B model. This task probes that performance enhancement.

Training No further training of DeepSeek-Prover-V2-7B is required of this task. In particular, we do not want to erase record of the sizable data set from DeepSeek-Prover-V2-7B’s weights.

Benchmarking Benchmarking will be against the test sets of miniF2F [179] and ProofNet [11] using pass@k metrics and proceed as in T1.1.

Timetable With DeepSeek-Prover-V2-7B being “dropped-in” as a replacement for the core DeepSeek-Prover-V1.5 model and requiring no further training, the work of this task is benchmarking. Hence, we budget 1 month for completion.

5.1.2.2 T2.2: Critic + DeepSeek-Prover-V2-7B Based MCTS Model

Research and Development As DeepSeek-Prover-V2-7B is a “drop-in” replacement for DeepSeek-Prover-V1.5, it can be dropped into the pipeline of T1.1. As this pipeline has already been researched and developed, no further work is required in this regard.

Training In contrast to T2.1, this task requires further training of the model. Here DeepSeek-Prover-V2-7B is being used (1) to produce Lean code and (2) to evaluate partial proofs, i.e. as a critic. It has been trained to produce Lean code, but it has not been trained as a critic. Hence, further training is required. This training, however, can proceed exactly as in T1.1 as T1.1 provided the exact same training regime but for DeepSeek-Prover-V1.5 instead of DeepSeek-Prover-V2-7B.

Benchmarking Benchmarking will be against the test sets of miniF2F [179] and ProofNet [11] using pass@k metrics and proceed as in T1.1.

Timetable With DeepSeek-Prover-V2-7B being “dropped-in” as a replacement for DeepSeek-Prover-V1.5, the core work will be training, which should require 1 month, and benchmarking, which we will allow an additional month for. Hence, we budget 2 months for completion.

5.1.2.3 T2.3: Alternative Bandit Algorithm + DeepSeek-Prover-V2-7B Based MCTS Model

Research and Development With DeepSeek-Prover-V2-7B being a “drop-in” replacement for DeepSeek-Prover-V1.5, we can drop it into the pipeline of T1.2. So, no further research and development is required.

Training No further training of DeepSeek-Prover-V2-7B is required. DeepSeek-Prover-V2-7B will be used in its normal capacity to produce Lean code and as the PUCT prior. Obviously, it requires no further training to be used in its normal capacity. It turns out when being used as a PUCT prior, it also requires no further training. When used as a prior, it functions normally but only its first proof step is retained to select the preferred child node.

Benchmarking Benchmarking will be against the test sets of miniF2F [179] and ProofNet [11] using pass@k metrics and proceed as in T1.2.

Timetable With the DeepSeek-Prover-V2-7B “dropped-in” and requiring no further training, the core task is benchmarking. For that we budget 1 month.

5.1.2.4 T2.4: Critic + Alternative Bandit Algorithm + DeepSeek-Prover-V2-7B Based MCTS Model

Research and Development Here we “drop-in” DeepSeek-Prover-V2-7B as a replacement for DeepSeek-Prover-V1.5 in the pipeline of T1.3. This requires no further research and development.

Training As in T2.2, this task requires the DeepSeek-Prover-V2-7B model be trained as a critic. This training will proceed as in T1.3.

Benchmarking Benchmarking will be against the test sets of miniF2F [179] and ProofNet [11] using pass@k metrics as in T1.3.

Timetable The core work is training, which should require about 1 month, and benchmarking, for which we will allow an additional month. Thus, we budget 2 months for this task.

5.1.2.5 T2.5-T2.11: Extended Training Times (Stretch Tasks)

Research and Development Extended training times generally lead to improved performance. However, this is particularly true for MCTS variants.

Training a MCTS variant necessarily involves expert iteration [4]. The raw model is trained on proofs produced from the enveloping, stronger MCTS variant. Hence, longer training times can lead to a marked increase in raw model performance. This can be thought of as a teacher/student setup where the teacher is the MCTS variant and the student is the raw model. But, as the MCTS variant leverages the raw model, improving the student also improves the teacher. A virtuous cycle.

Existence of this virtuous cycle implies that MCTS variants experience particularly marked performance gains from extend training times. These stretch tasks explore these performance gains.

For each existing task T1.1-T2.4 this task introduces a corresponding, new task T2.5-T2.11 that introduces or extends the training time of its partner. For example, the task T2.5 will correspond to the task T1.1, having exactly the same setup, but will extend the training time of T1.1, tracking validation performance on miniF2F and ProofNet to monitor any performance increase or to end training when no further performance increases occur.

Training If a task’s partner, e.g. T1.1, has a training regime, then the new task, e.g. T2.5, will follow that regime. If a task’s partner, e.g. T2.1, has no training regime, then the new task, e.g. T2.8, introduces an appropriate training regime, e.g. the DeepSeek-Prover-V1.5 training regime.

Benchmarking Benchmarking of each new task, e.g. T2.5, follows the benchmarking procedures of its partner, e.g. T1.1.

Timetable As each of these new tasks involves only training and benchmarking, we will allocate 2 months for each.

5.2 Formalizing Algebraic Quantum Field Theory in Lean

The three formalizing work packages—Variant Formalization (WP3), Foundational Result Formalization (WP4), and Advanced Result Formalization (WP5)—describe our approach to the research questions of subsection 3.2 Formalizing Algebraic Quantum Field Theory. Variant Formalization (WP3) lays out the details of our plan for attacking questions raised in paragraph 3.2.2.1 Haag–Kastler Variants. Questions on foundational results posed in paragraph 3.2.2.2 Foundational Results are covered in Foundational Result Formalization (WP4), and research questions on advanced results from paragraph 3.2.2.2 Foundational Results are taken up in Advanced Result Formalization (WP5).

5.2.1 WP3: Variant Formalization

The original Haag-Kastler axioms [64] have, over time, given rise to variants—subtle and not so subtle revisions of the originals meant to clarify and/or modify Haag and Kastler’s work. These variants take many forms. For example, a variant may base its net of C^* -algebras not on bounded open sets but on causally complete sets⁴¹. Alternatively, a variant may not require a representation of the Poincaré group on its quasi-local algebra but may only require a representation of the translation subgroup. Possibilities for variants are endless.

These possibilities, however, precipitate problems. How are all these variants related? What’s their web of entailment? Which are equivalent? Which are inequivalent? The mere existence of these variants muddies the waters; it’s unclear how the variants relate to one another, much less to the real world.

This work package clarifies this murk. Formalizing variants, it crystallizes variants’ axioms. Establishing entailments, it methodically untangles variants’ web of relations. Exhibiting equivalence, it establishes which variants are simply distinctions without a difference.

5.2.1.1 T3.1: Formalization The objective of this task is to formalize Haag-Kastler variants in Lean. At a minimum the five “most popular” variants will be formalized. These are:

- **Haag-Kastler Variant** - Original Haag-Kastler axioms [64] from 1964.
- **Haag Variant** - Variant appearing in the second edition of Haag’s *Local Quantum Physics: Fields, Particles, Algebras* [63] from 1996.
- **Araki Variant** - Variant from the English edition of Araki’s *Mathematical Theory of Quantum Fields* [5] published in 1999.
- **Borchers Variant** - Variant of Borchers’ in the published version of *On revolutionizing quantum field theory with Tomita’s modular theory* [18] from 2000.
- **Fredenhagen Variant** - Variant appearing in Fredenhagen’s *An Introduction to Algebraic Quantum Field Theory* [55] published in 2015.

Research necessitated to complete this task may uncover additional variants that warrant formalization. The decision to formalize or not to formalize these additional variants will be made on a case-by-case basis. That being said, these five are the core of this task’s effort.

Lean Library This task will be the first to produce Lean code which is to reside in our library physicslib. Hence, step zero, before producing any Lean code, is to create physicslib itself. We will do so using lake [87] or the LeanProject template [115]. physicslib will reside under the physicslib

⁴¹The casual complement $\mathcal{O}'$ of a set $\mathcal{O}$ in Minkowski space is the set of points space-like related to all points of $\mathcal{O}$. A set $\mathcal{O}$ is casually complete if $\mathcal{O} = \mathcal{O}''$.

organization in GitHub. This GitHub organization will be the “home” for physicslib and the work of this project.

Lean Project Larger Lean projects tend to use leanblueprint [114] to create a project organization framework. We won’t break that mold. leanblueprint will be called upon to create such a framework along with a blueprint in the GitHub repository physicslib/physicslib. This blueprint will outline formalization of the Haag-Kastler variants.

Formalization For each variant, formalization proper begins by drafting a sketch of the variant’s axioms. Sketches of all the non-existent dependencies of this sketch are then sketched, then sketches of all non-existent dependencies of those sketches are sketched... This process is repeated until all required dependencies have either been sketched or imported from, say, mathlib. In this manner a variant’s blueprint is “fleshed out”. This is unfortunately a time consuming process.

But, we can speed all this up. Motivated by DeepSeek-Prover-V2 [119] we can employ the open source model DeepSeek-V3 to autoformalize variants’ axioms. These will provide quick sketches that we (or DeepSeek-V3) can build upon. For example, simply prompting DeepSeek-V3’s with⁴²:

In the context of the Haag-Kastler axioms of algebraic quantum field theory, can you formalize this statement of the isotony axiom in the Lean programming language?

Let $\{\mathcal{O}_i\}$ be the set of all open sets of Minkowski spacetime with compact closure. If $\mathcal{O}_1$ and $\mathcal{O}_2$ in $\{\mathcal{O}_i\}$ satisfy $\mathcal{O}_1 \subset \mathcal{O}_2$, then their abstract C^* -algebras satisfy $\mathcal{C}(\mathcal{O}_1) \subset \mathcal{C}(\mathcal{O}_2)$, where $\mathcal{C}(\mathcal{O}_1)$ and $\mathcal{C}(\mathcal{O}_2)$ have either a common unit or both have no unit.

yielded in under a minute this sketch of the isotony axiom

```
import Mathlib.Topology.Basic
import Mathlib.Topology.SubsetProperties
import Mathlib.Algebra.Star.Basic
import Mathlib.Analysis.NormedSpace.StarAlgebras

open Set

-- We assume Minkowski spacetime as a topological space
variable (M : Type*) [TopologicalSpace M] [LocallyCompactSpace M]

-- The set of open sets with compact closure
def openSetsWithCompactClosure : Set (Set M) :=
  {O : Set M | IsOpen O ∧ IsCompact (closure O)}

-- A  $C^*$ -algebra structure (simplified representation)
class CStarAlgebra (α : Type*) extends StarRing α, NormedAddCommGroup α, NormedSpace ℂ α :=
  (norm_mul : ∀ a b, ||a * b|| ≤ ||a|| * ||b||)
  (norm_star : ∀ a, ||star a|| = ||a||)
  (cstar_property : ∀ a, ||star a * a|| = ||a|| * ||a||)

-- The net of  $C^*$ -algebras indexed by open sets with compact closure
variable {C : Set M → Type*} [∀ O, CStarAlgebra (C O)]

-- The isotony axiom
```

⁴²DeepSeek-V3 was prompted with the “raw” LaTeX of the following.

```

axiom isotony :  $\forall (O_1 O_2 : \text{Set } M),$ 
   $O_1 \in \text{openSetsWithCompactClosure } M \rightarrow$ 
   $O_2 \in \text{openSetsWithCompactClosure } M \rightarrow$ 
   $O_1 \subseteq O_2 \rightarrow$ 
   $(\exists (\text{embed} : C O_1 \rightarrow_{\star} [\mathbb{C}] C O_2), \text{Function.Injective embed}) \wedge$ 
   $((\exists (1_1 : C O_1), \forall a, 1_1 \star a = a \wedge a \star 1_1 = a) \leftrightarrow$ 
   $(\exists (1_2 : C O_2), \forall a, 1_2 \star a = a \wedge a \star 1_2 = a))$ 

```

Not perfect, but a reasonable starting point. One can pose followup prompts to refine this sketch, e.g. specify use of the standard C^* -algebra definition, or simply refine it manually.

Beyond formalizing missing definitions, e.g. adding the definition of C^* -inductive limit, “fleshing out” the blueprint can involve formalizing and then proving propositions, e.g. formalizing the statement of then proving that the C^* -inductive limit is a C^* -algebra. Manually formalizing and proving these propositions is a time sink.

However, we built an IPA to accelerate all this. (1) Our IPA can be given a chance to generate a proposition proof outright. (2) If that fails, DeepSeek-V2-7B can be prompted to create one or more formalized proposition proof blueprints. (3) Our IPA can then be given the opportunity to generate a proof for all lemmas in each blueprint. (4) For lemmas it can’t prove, it’s given human assistance, via proof step generation, for each unproven lemma.

With this deep learning accelerates the entire process. Autoformalization, proof generation, and proof step generation in concert hasten all phases involved.

Timeline With the accelerants of DeepSeek-V3, DeepSeek-V2-7B, and our IPA, formalization should proceed rapidly. We estimate that formalization of all variants along with the associated definitions, propositions, and lemmas should take a total of 2 months.

5.2.1.2 T3.2: Entailment The objective of this task is to formalize proofs of all possible variant entailment relations in Lean.

Explicitly, the axioms of one variant are “demoted” to theorems, then an attempt is made to use the axioms of a second variant to formally prove these theorems. For example, the isotony axiom of the Haag-Kastler variant is “demoted” to a theorem, then the axioms of, say, the Araki variant are used to formally prove this theorem. If all “demoted” theorems of the Haag-Kastler variant can be formally proven using the axioms of the Araki variant, then the Haag-Kastler variant is entailed by the Araki variant. Similar arguments used for all variant pairs yield all possible variant entailment relations.

Establishing entailment can be time consuming. In task T3.1 we will have formalized 5-10 different variants each with 5-10 axioms. Thus, as we have previously seen, establishing all variant entailment relations requires 20 to 900 formal proofs. Completing 900 formal proofs manually is a daunting task. Luckily, we built an IPA to accelerate all this.

However, instead of turning our IPA loose on all variants simultaneously, we will establish entailment pairwise to minimize (human) context switching, i.e. considering a pair of variants then determining if the first entails the second and subsequently determining if the second entails the first.

Theorem Targeted search_proof First we will deploy `search_proof`⁴³ in an attempt to establish variant entailment.

Explicitly, given a pair of variants we “demote” all axioms of the first variant to theorems, then prompt `search_proof` to prove these theorems using the axioms from the second variant. This will

⁴³For illustration purposes we use the LeanCopilot [131] constructs `search_proof` and `suggest_tactics` here and below. Equivalently, we just as well could have used the analogous constructs from Pantograph [3] instead.

result in some of these theorems being proven, and some remaining unproven.

Proof Blueprint Formalization For the unproven theorems we manually create a formal sketch of a proof using the axioms of the second variant. Alternatively, we prompt DeepSeek-V2-7B to create such a sketch. Either way we end up with a formal proof sketch using the axioms of the second variant. This sketch will contain propositions with sorry proof placeholders.

Proposition targeted search_proof For each unproven proposition in the formal proof sketch, we prompt search_proof to prove it using the axioms from the second variant. This results in some being proven and some left unproven.

Proposition targeted suggest_tactics For the unproven propositions we manually walk through their proofs utilizing suggest_tactics along the way to speed progress.

If all “demoted” theorems from the first variant are proven using the axioms of the second variant, we establish entailment, the second entailing the first. If not, we then switch the roles of the first and second variants and repeat the process above. This entire process is then repeated for each pair of variants.

Timeline It’s difficult to establish a reliable time estimate for this task. Such an estimate relies critically upon the “strength” of our IPA which is, as of yet, unknown. However, we can establish upper and lower bounds.

Assuming a superhuman IPA, it could immediately prove or disprove each “demoted” theorem when search_proof is first prompted⁴⁴. This would allow for all entailment relations to be established within a couple of days at worst.

Assuming the IPA is unable to help in any regard and all proofs are manual, this task could take from a month to a year for 20-900 “demoted” theorems.

Splitting the difference, we estimate this task will take 6 months. This will likely be a conservative estimate as, in parallel to our work on our IPA, DeepSeek will be putting all effort into improving and open sourcing the successor of DeepSeek-Prover-V2. This successor will doubtlessly outperform DeepSeek-Prover-V2 and could be integrated into Lean and used in our toolchain, improving upon our IPA and accelerating the process.

5.2.1.3 T3.3: Equivalence The objective of this task is to formalize proofs of all possible variant equivalence relations in Lean.

Explicitly, if one variant is entailed by a second and the second is entailed by the first, then the variants are equivalent. As task T3.2 establishes all variant entailment relations, this task simply amounts to collating these results into equivalence relations.

Timeline As the vast majority of the work for this task will be completed in task T3.2, this task requires minimal time. Our minimal task time resolution is one month. So we allocate 1 month for this task.

5.2.2 WP4: Foundational Result Formalization

Algebraic quantum field theory is more than a collection of foundational axioms. Its has superstructure built upon substructure built in turn upon the foundational axioms. Thus, its substructure supports

⁴⁴We ignore incompleteness.

all advanced quantum field theory. GNS construction, Reeh-Schlieder theorem, translation invariance of the vacuum, and scores of other results are testament to this substructure’s importance. Therefore, formalizing substructure is critical.

However, this substructure is often carelessly constructed. For example, in Haag’s *Local Quantum Physics: Fields, Particles, Algebras* [63] the Reeh-Schlieder theorem is *not* proven using Haag’s Variant, but it is proven using Wightman’s Axioms [133]. Despite this, superstructure, claimed to be founded upon Haag’s Variant, is built upon it [63]. This is unacceptable.

This work package renovates this substructure. Formalizing statements of these foundational results, it clarifies the substructure’s definitions, theorems, propositions, and lemmas. Proving these foundational results, it formalizes proofs of the substructure’s key theorems.

5.2.2.1 T4.1: Formalization The objective of this task is to formalize the foundational results⁴⁵ of algebraic quantum field theory in Lean. These results should include, but not be limited to:

- *Theorem 1 (GNS Construction)* - Proving products of the GNS Construction, e.g. the Hilbert space $\mathcal{H}_\omega$, have the promised properties [58].
- *Theorem 2.1.4 (Properties of von Neumann Algebras)* - Establishing various von Neumann algebra properties [63].
- *Theorem 4.5 (Translation Invariance of Vacua)* - Establishing vacua are translation invariant [5].
- *Theorem 4.6.2 (Properties of the Vacua)* - Proving standard, fundamental properties of vacua [5].
- *Theorem 4.7 (Mass Gap and Cluster Decomposition)* - Proving existence of a mass gap implies the cluster decomposition property holds [5].
- *Theorem 4.11 (Truncated Cluster Decomposition)* - Proving a cluster decomposition property holds for truncated correlation functions [5].
- *Theorem 3.2.2 (Primary States and Cluster Decomposition)* - Proving primary states exhibit the cluster decomposition property [63].
- *Theorem 4.14 (Reeh-Schlieder theorem)* - Exhibiting the Reeh-Schlieder theorem [5] without recourse to Wightman’s Axioms [133].

Formalization of these foundational results requires the definitions, propositions, and lemmas they are based upon also be formalized. This is a non-trivial task and may involve the bulk of the work.

Also note that some of these formalizations may end up in `mathlib` as opposed to `physicslib`. For example, both *Theorem 1 (GNS Construction)* and *Theorem 2.1.4 (Von Neumann Algebra Properties)* are more apropos for `mathlib` than `physicslib`.

Lean Project For each “top-level” theorem formalized, e.g. *Theorem 1 (GNS Construction)*, we will employ `leanblueprint` [114] to create a blueprint and project organization framework. The blueprint will organize formalization of the “top-level” theorem.

⁴⁵Foundational results will refer to definitions, theorems, postulates, and lemmas covered in, but not limited to, the following references : Chapter III of Haag’s *Local Quantum Physics: Fields, Particles, Algebras* [63] and Chapter IV of Araki’s *Mathematical Theory of Quantum Fields* [5].

Formalization As in task T3.1, formalization begins with a sketch-formalization of a “top-level” theorem’s statement, but not its proof. This is followed with sketches of all the non-existent dependencies, sketches of the sketches’ dependencies... This process is repeated until all required dependencies have been sketched or imported from, say, `mathlib`, fully “fleshing out” the blueprint. This process can be time consuming.

However, as in T3.1, DeepSeek-V3, DeepSeek-V2-7B, and our IPA can accelerate all steps involved. DeepSeek-V3 can be employed to autoformalize statements of “top-level” theorems and their dependencies. When these dependencies are theorems, propositions, or lemmas that require formal proof, we can exploit our IPA along with DeepSeek-V2-7B to accelerate proof formalization⁴⁶.

Timeline With the help of DeepSeek-V3, DeepSeek-V2-7B, and our IPA, we should proceed rapidly. However, the sheer number of “top-level” theorems and dependencies is massive. That said, we estimate statement formalization of all “top-level” theorems and dependencies (which can include formal proofs) to take 3 months.

5.2.2.2 T4.2: Proof Generation The objective of this task is to formalize proofs of all foundational theorems under all variant equivalence classes in Lean. That being said, one or more foundational theorems may not follow from the axioms of a particular variant equivalence class. Hence, our objective is more accurately stated as follows: Formalize proofs of all foundational theorems under as many variant equivalence classes as possible in Lean. Considering the number of variant equivalence classes and the number of foundational theorems, this task is indeed daunting. However, as in task T3.2, we will enlist DeepSeek-V2-7B and our IPA to accelerate this work.

Theorem Targeted search_proof As in T3.2, we first deploy `search_proof` in attempt to prove each “top-level” theorem; it’s deployed once for each variant equivalence class. This will result in some “top level” theorems being proven, and some remaining unproven.

Proof Blueprint Formalization For each unproven theorem we manually create a formal proof sketch for each variant equivalence class, or we prompt DeepSeek-V2-7B to create such sketches. Either path leads to a set of formal proof sketches containing unproven propositions with sorry placeholders.

Proposition Targeted search_proof For each unproven proposition in the formal proof sketch, we prompt `search_proof` to prove it using the apropos variant equivalence class. Some propositions will succumb. However, others will remain unproven.

Proposition Targeted suggest_tactics For those that remain unproven, we manually walk our IPA through a proof using `suggest_tactics`.

Timeline It’s difficult to establish an exact timeline as some fundamental results may simply be false for a particular variant equivalence class. However, with the experience gained up until this point in completing other proofs, we estimate it will take 3 months to complete this task.

5.2.3 WP5: Advanced Result Formalization (Stretch Objective)

Built upon the substructure we establish in WP4 is the superstructure of algebraic quantum field theory. This superstructure is diverse and sizable, covering topics from the DHR-Analysis [63] to the quantum null-energy condition [12]. We could never hope to cover it all.

⁴⁶We must take care to establish each such result for each variant equivalence class when possible.

Thus, this work package covers aspects of this superstructure built directly upon the substructure we established in WP4. Formalizing statements of these advanced results, it clarifies associated definitions, theorems, propositions, and lemmas. Proving these advanced results, it verifies their soundness.

5.2.3.1 T5.1: Formalization The objective of this task is to formalize the advanced results of algebraic quantum field theory in Lean. These results should include, but not be limited to⁴⁷:

- *Corollary 3.13 (Tomita's Theorem)* - Proving the modular automorphism maps a von Neumann algebra to itself [132].
- *Theorem V.4.1.1 (Bisognano Wichmann Theorem)* - Proving, within the confines of algebraic quantum field theory, the Bisognano Wichmann Theorem [63, 100].

As in task T4.1, formalization requires all definitions, propositions, and lemmas these theorems rely on also be formalized. It's likely this will be the bulk of the work.

As in T4.1, this task creates a Lean project for each "top-level" theorem and executes subsequent formalization work under this project. The steps involved are the same as those in T4.1 but operate on this different set of "top-level" theorems. So, one can refer to T4.1 for details.

Timeline Assisted by DeepSeek-V3, DeepSeek-V2-7B, and our IPA, rapid progress can be made. However, the number of "top-level" theorems and dependencies hinders rapid completion. We estimate statement formalization of all "top-level" theorems and dependencies (which can include formal proofs) to take 4 months.

5.2.3.2 T5.2: Proof Generation The objective of this task is to formalize proofs of all advanced theorems under as many variant equivalence classes as possible in Lean. As in T4.2, this will involve significant work. So we enlist our IPA as an assistant.

The steps involved here are as in task T4.2: *Theorem Targeted search_proof*, *Proof Blueprint Formalization*, *Proposition Targeted search_proof*, and *Proposition Targeted suggest_tactics*. Hence, one can refer to T4.2 for the details.

Timeline With the rapid advances of deep learning, there's considerable difficulty projecting this far into the future. Models may have advanced to the point where the proofs here are trivial, or they may be stumped by such proofs. Either way, we estimate it will take roughly 4 months to complete this task.

5.3 Dissemination and Exploitation

Without dissemination and exploitation, all work package results are useless. Using deep learning, the formalization work packages may definitively establish, say, the AdS/CFT correspondence, but without dissemination the Lean, physics, and deep learning communities will be none-the-wiser. Theorem proving work packages may produce models capable superhuman feats, but if they remain unknown, they remain unexploited and are of no import. Dissemination and exploitation are critical to project success.

Thus, this work package ensures the results of all work packages are disseminated to and exploited by experts and neophytes alike. Using open source strategies, blogging, discussion forums, social media, conferences, and academic publishing it accelerates dissemination and exploitation of all work package results.

⁴⁷This is a stretch objective. Hence, this list is subject to change.

5.3.1 WP6: Dissemination and Exploitation

The objective of this work package is to build a large, diverse, vibrant community around SotA open models, Lean integration, and formalization to trigger a sea change in the way physics is done.

The strategy to achieve this objective encompasses many disparate goals: (1) open sourcing models and formalization boosts community, building and expanding exploitation; (2) productizing models' Lean integration bends neophytes' learning curve, enlarging community; (3) cultivating vibrant, online forums fosters community; (4) creating and maintaining an active website and social media footprint strengthens community and cultivates its growth; (5) presenting at scholarly conferences and publishing in academic journals promotes expert participation. A diverse set of strategies.

This diverse set of strategies is required to reach a diverse audience. Our project targets Lean experts with little physics or deep learning experience, physicists with vanishing Lean and/or deep learning experience, deep learning experts with only passing knowledge of physics or Lean, and complete neophytes to all disciplines involved. This legion is ill-served by one, or even a few strategies. A manifold of strategies is mandatory.

5.3.1.1 T6.1: Website Creation and Maintenance The objective of this task is to create and maintain a project website. This serves many strategic goals.

Communities across all audiences will benefit, e.g. regular blog posts targeting experts or neophytes bring all audiences into the fold. Technical blog posts marking milestones, e.g. publication of a new academic article, and targeting experts are de rigueur for many disciplines. We will not waver from this standard. In contrast, blog posts targeting neophytes are a perfect match for the web as there is no established place for such in academic publishing.

Our website will also inform internal and external stakeholders about the project's progress. Hosting links to the project's blueprints, it will serve as a hub from which one can find the status of any particular work package or task. In addition, blog posts announcing releases of code, documentation, and/or models will be centralized on the website and further promoted there. Notices of upcoming conference or workshop presentations will be centralized here and promoted through social media, buoying the website's importance and reach. Past conference or workshop presentations will be hosted on the website too, allowing those unable to attend to still benefit. Our website will also serve as a hub to access all academic articles produced by the project.

Timeline Website creation will take place in parallel with T1.1 during the first 3 months of the project. To minimize the work required, we will avail ourselves of GitHub pages to manage the website. Upon the completion of each milestone, e.g. completion of WP1, a blog post or posts will be created covering accomplishments of the milestone. In addition, as previously mentioned, intermittently other information will be posted too, e.g. notices of upcoming conference or workshop presentations.

5.3.1.2 T6.2: Productization of Lean Integration The objective of this task is to productize the model's Lean integration. Several strategic goals are addressed through this objective.

Neophytes to deep learning, Lean, and/or physics will not want to spend an inordinate amount of time installing our model, finding out how to integrate it into Lean, worrying about shared objects, or any other number of minutia. They want to install a library and have it work.

Achieving this requires not only that the model is integrated into Lean, but that all the "rough edges" of the model be "sanded" away. For example, documentation should be present, clear, and current, and model access shouldn't require knowing the ins and outs of GitHub releases or Hugging Face. It should just work. Achieving this productization is the objective of this task.

Productization involves a myriad of subtasks. Documentation must be created and maintained. Remote and local model execution must be created, productized, and maintained. Problematic edge cases of package installation must be identified and dispatched. This gives the flavor of the tasks involved, but numerous other tasks also fall under this umbrella.

Productization is critical to enlarging and diversifying community. Ease of use allows potential community members to “kick-the-tires” while they’re still excited about the project’s prospects, increasing the chances they’ll join the community instead of moving on frustrated with endless yak shaving. This applies to neophytes and experts alike. Just because someone is a deep learning expert and is able to decipher any PyTorch problem doesn’t mean they want to.

Timeline After the completion of WP1 and WP2, productization of the lean integration will occur. This will occur in parallel with other tasks and therefore take 2 months.

5.3.1.3 T6.3: Fostering Communities The objective of this task is to foster communities across all our audiences. Almost all strategic goals are addressed here.

Subtasks involved here run the gamut. Open sourcing our model(s) and formalization grows our community bringing in open source enthusiasts. Productization of Lean Integration (T6.2) also serves this objective, expanding our community with neophytes and experts. Creating, growing, and maintaining online forums dedicated to our project plays a key role in fostering community; online forums provide immediate and daily feedback which isn’t true of most other channels. Blogging, part of T6.1, also serves this goal, gently on-boarding new users while also providing deep dives for the more experienced. Task T6.4, presenting at scholarly conferences and publishing in academic journals, is also in service of this goal; it fosters an academic community. Finally, social media outreach maintains community through frequent updates and grows community through simple serendipity.

Timeline This process is ongoing throughout the entire project, e.g. we will always be interacting with members of our online forums.

5.3.1.4 T6.4: Conferences and Publications The objective of this task is to present at apropos conferences and publish in appropriate journals. Generally this targets strategic goals that have expert audiences.

Expert communities congregate around specific conferences and journals. Lean has its set of events [88], e.g. the Interactive Theorem Proving 2025 conference. Physics has its events [75], e.g. the QIQG 2025: Quantum Information in Quantum Gravity conference. While deep learning has its separate set of events [112], e.g. the NeurIPS 2025 conference. Similarly, each community has its own preferred journals.

This task naturally breaks into a number of subtasks: (1) ascertaining appropriate conference deadlines that align with project milestones; (2) planning submission of articles to those appropriate, aligning conferences with the goal of lecturing there; (3) writing up the results; and (4) submitting related journal articles to appropriate journals contemporaneously with the conferences.

Timeline These subtasks will occur after milestone completions, but exactly when is dependent upon conference scheduling.

5.3.1.5 T6.5: Collaborations The objective of this task is to foster direct, expert collaboration with universities and/or research groups. This targets strategic goals with expert audiences.

With the diversity of topics—Lean, deep learning, and physics—there will always be someone somewhere with expert knowledge we simply do not have. We can leverage this to foster expert communities.

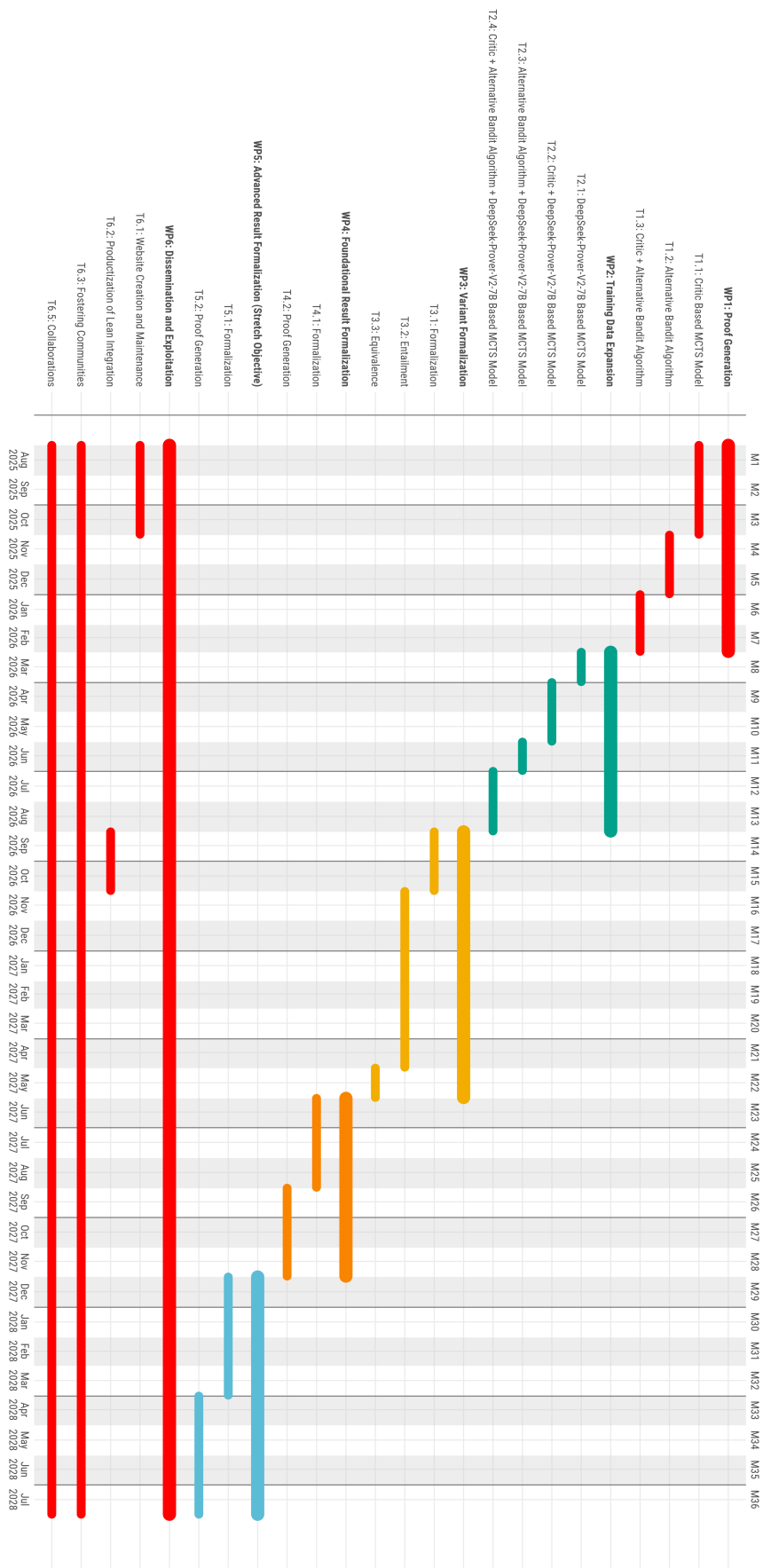
Experts are curious. So, if we come to someone in the Princeton Language and Intelligence group, say, with a deep learning question, surely they can provide an answer. But, the question alone might not pique their curiosity. However, coupling the question with, say, its algebraic quantum field theory context would peak their curiosity and draw them in, making it more likely they answer and more likely they're intrigued and want to stay in touch.

There are many researchers and groups we plan to reach out to; they include:

- Sanjeev Arora(Princeton) - Deep learning and Lean
- Klaus Fredenhagen(Universität Hamburg) - Algebraic quantum field theory
- Stefan Hollands(Universität Leipzig) - Algebraic quantum field theory
- Floris van Doorn(Universität Bonn) - Lean
- Igor Khavkine(Czech Academy of Sciences) - Algebraic quantum theory and Lean
- Yoh Tanimoto(Università di Roma) - Algebraic quantum theory and Lean
- Frédéric Dupuis(University of Montreal) - Operator algebras and Lean
- Anatole Dedecker(University of Paris Diderot) - Operator algebras and Lean
- Jireh Loreaux(Southern Illinois University) - Operator algebras and Lean
- Kim Morrison(Lean FRO) - Operator algebras and Lean
- Sebastian Pokutta and Christoph Spiegel(ZIB) - Lean

Timeline This process is ongoing throughout the project.

6 Timetable



7 Bibliography

- [1] B C. Allanach, Ben Gripaios, and Joseph Tooby-Smith. Geometric general solution to the $U(1)$ anomaly equations. *Journal of High Energy Physics*, 2020(5), May 2020. ISSN 1029-8479. doi: 10.1007/jhep05(2020)065. URL [http://dx.doi.org/10.1007/JHEP05\(2020\)065](http://dx.doi.org/10.1007/JHEP05(2020)065).
- [2] Chenyang An, Zhibo Chen, Qihao Ye, Emily First, Letian Peng, Jiayun Zhang, Zihan Wang, Sorin Lerner, and Jingbo Shang. Learn from Failure: Fine-Tuning LLMs with Trial-and-Error Data for Intuitionistic Propositional Logic Proving. *arXiv preprint arXiv:2404.07382*, 2024.
- [3] Leni Aniva, Chuyue Sun, Brando Miranda, Clark Barrett, and Sanmi Koyejo. Pantograph: A Machine-to-Machine Interaction Interface for Advanced Theorem Proving, High Level Reasoning, and Data Extraction in Lean 4, 2025. URL <https://arxiv.org/abs/2410.16429>.
- [4] Thomas Anthony, Zheng Tian, and David Barber. Thinking Fast and Slow with Deep Learning and Tree Search, 2017. URL <https://arxiv.org/abs/1705.08439>.
- [5] Huzihiro Araki. *Mathematical Theory of Quantum Fields*. International series of monographs on physics 101, Pr.2009. Oxford University Press, Oxford, [reprinted] edition, 2009. ISBN 9780199566402.
- [6] V.I. Arnold. *Mathematical methods of classical mechanics*, volume 60. Springer, 1989.
- [7] Michael F Atiyah. Topological quantum field theory. *Publications Mathématiques de l’IHÉS*, 68: 175–186, 1988.
- [8] Peter Auer, Nicolo Cesa-Bianchi, and Paul Fischer. Finite-time analysis of the multiarmed bandit problem. *Machine learning*, 47:235–256, 2002.
- [9] Eser Aygün, Zafarali Ahmed, Ankit Anand, Vlad Firoiu, Xavier Glorot, Laurent Orseau, Doina Precup, and Shibl Mourad. Learning to Prove from Synthetic Theorems. *arXiv preprint arXiv:2006.11259*, 2020.
- [10] Eser Aygün, Ankit Anand, Laurent Orseau, Xavier Glorot, Stephen M McAleer, Vlad Firoiu, Lei M Zhang, Doina Precup, and Shibl Mourad. Proving Theorems using Incremental Learning and Hindsight Experience Replay. In *Proceedings of the International Conference on Machine Learning*, 2022.
- [11] Zhangir Azerbayev, Bartosz Piotrowski, Hailey Schoelkopf, Edward W. Ayers, Dragomir Radev, and Jeremy Avigad. ProofNet: Autoformalizing and Formally Proving Undergraduate-Level Mathematics, 2023. URL <https://arxiv.org/abs/2302.12433>.
- [12] Srivatsan Balakrishnan, Thomas Faulkner, Zuhair U. Khandker, and Huajia Wang. A general proof of the quantum null energy condition. *Journal of High Energy Physics*, 2019(9), September 2019. ISSN 1029-8479. doi: 10.1007/jhep09(2019)020. URL [http://dx.doi.org/10.1007/JHEP09\(2019\)020](http://dx.doi.org/10.1007/JHEP09(2019)020).
- [13] Kshitij Bansal, Sarah M. Loos, Markus Norman Rabe, Christian Szegedy, and Stewart Wilcox. HOList: An Environment for Machine Learning of Higher Order Logic Theorem Proving. In *Proceedings of the International Conference on Machine Learning*, 2019.
- [14] Filip Bárték and Martin Suda. How Much Should This Symbol Weigh? A GNN-Advised Clause Selection. In *Proceedings of International Conference on Logic for Programming, Artificial Intelligence and Reasoning*, 2023.

- [15] Andrej Bauer, Matej Petković, and Ljupco Todorovski. MLFMF: Data Sets for Machine Learning for Mathematical Formalization. In *Proceedings of the International Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023.
- [16] Christian Blanchet and Vladimir Turaev. Axiomatic approach to topological quantum field theory. *Encyclopedia of mathematical physics*. Academic Press/Elsevier Science, Oxford, 2005.
- [17] N. N. Bogolyubov, A. A. Logunov, A. I. Oksak, and I. T. Todorov. *General principles of quantum field theory*. Mathematical Physics and Applied Mathematics. Kluwer Academic Publishers, 1990.
- [18] H. J. Borchers. On revolutionizing quantum field theory with Tomita’s modular theory. *J. Math. Phys.*, 41:3604–3673, 2000. doi: 10.1063/1.533323.
- [19] H. J. Borchers and J. Yngvason. On the PCT–Theorem in the Theory of Local Observables, 2000. URL <https://arxiv.org/abs/math-ph/0012020>.
- [20] Raphael Bousso, Zachary Fisher, Jason Koeller, Stefan Leichenauer, and Aron C. Wall. Proof of the quantum null energy condition. *Physical Review D*, 93(2), January 2016. ISSN 2470-0029. doi: 10.1103/physrevd.93.024017. URL <http://dx.doi.org/10.1103/PhysRevD.93.024017>.
- [21] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners, 2020. URL <https://arxiv.org/abs/2005.14165>.
- [22] Detlev Buchholz and Klaus Fredenhagen. Algebraic quantum field theory: objectives, methods, and results, 2024. URL <https://arxiv.org/abs/2305.12923>.
- [23] Detlev Buchholz, Sergio Doplicher, Roberto Longo, and John E. Roberts. A New look at Goldstone’s theorem. *Rev. Math. Phys.*, 4(spec01):49–83, 1992. doi: 10.1142/S0129055X92000157.
- [24] Kevin Buzzard. Formalising mathematics: an introduction, 2021. URL <https://xenaproject.wordpress.com/2021/01/21/formalising-mathematics-an-introduction/>.
- [25] Kevin Buzzard. What is the point of computers? A question for pure mathematicians, 2022. URL <https://arxiv.org/abs/2112.11598>.
- [26] Kevin Buzzard, Johan Commelin, and Patrick Massot. Formalising perfectoid spaces. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, page 299–312. ACM, January 2020. doi: 10.1145/3372885.3373830. URL <http://dx.doi.org/10.1145/3372885.3373830>.
- [27] Kevin Buzzard, Chris Hughes, Kenny Lau, Amelia Livingston, Ramon Fernández Mir, and Scott Morrison. Schemes in Lean. *Experimental Mathematics*, 31(2):355–363, November 2021. ISSN 1944-950X. doi: 10.1080/10586458.2021.1983489. URL <http://dx.doi.org/10.1080/10586458.2021.1983489>.
- [28] Ewen Callaway. ‘It will change everything’: DeepMind’s AI makes gigantic leap in solving protein structures. *Nature*, 588(7837):203–205, 2020.

- [29] Edresson Casanova, Kelly Davis, Eren Gölge, Görkem Gökner, Iulian Gulea, Logan Hart, Aya Aljafari, Joshua Meyer, Reuben Morais, Samuel Olayemi, and Julian Weber. XTTS: a Massively Multilingual Zero-Shot Text-to-Speech Model, 2024. URL <https://arxiv.org/abs/2406.04904>.
- [30] Fikret Ceyhan and Thomas Faulkner. Recovering the QNEC from the ANEC, 2019. URL <https://arxiv.org/abs/1812.04683>.
- [31] Venkatesa Chandrasekaran, Roberto Longo, Geoff Penington, and Edward Witten. An algebra of observables for de Sitter space. *Journal of High Energy Physics*, 2023(2), February 2023. ISSN 1029-8479. doi: 10.1007/jhep02(2023)082. URL [http://dx.doi.org/10.1007/JHEP02\(2023\)082](http://dx.doi.org/10.1007/JHEP02(2023)082).
- [32] Charlie Wood. If the Universe Is a Hologram, This Long-Forgotten Math Could Decode It, 2024. URL <https://shorturl.at/9lmw0>.
- [33] Kyunghyun Cho, Bart Van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 2014.
- [34] Karel Chvalovský, Jan Jakubův, Martin Suda, and Josef Urban. ENIGMA-NG: Efficient Neural and Gradient-Boosted Inference Guidance for E. In *Proceedings of the International Conference on Automated Deduction*, 2019.
- [35] Karel Chvalovský, Jan Jakubův, Miroslav Olšák, and Josef Urban. Learning Theorem Proving Components. In *Proceedings of the International Conference on Automated Reasoning with Analytic Tableaux and Related Methods*, 2021.
- [36] common voice. common-voice/common-voice. <https://github.com/common-voice/common-voice>, 2025.
- [37] Coqui.ai Corp. coqui-ai/STT. <https://github.com/coqui-ai/STT>, 2025.
- [38] Coqui.ai Corp. coqui-ai/TTS. <https://github.com/coqui-ai/TTS>, 2025.
- [39] Davi B. Costa, Bogdan A. Dobrescu, and Patrick J. Fox. General Solution to the $U(1)$ Anomaly Equations. *Physical Review Letters*, 123(15), October 2019. ISSN 1079-7114. doi: 10.1103/physrevlett.123.151601. URL <http://dx.doi.org/10.1103/PhysRevLett.123.151601>.
- [40] Rémi Coulom. Efficient selectivity and backup operators in Monte-Carlo tree search. In *International conference on computers and games*, pages 72–83. Springer, 2006.
- [41] Rémi Coulom. Computing Elo ratings of move patterns in the game of Go. *ICGA Journal*, 30(4): 198–208, 2007. doi: 10.3233/ICG-2007-30403. URL <https://journals.sagepub.com/doi/abs/10.3233/ICG-2007-30403>.
- [42] Doratossadat Dastgheib and Ehsaneddin Asgari. Keyword-Based Natural Language Premise Selection for an Automatic Mathematical Statement Proving. In *Proceedings of TextGraphs-16: Graph-Based Methods for Natural Language Processing*, 2022.
- [43] P C W Davies. Scalar production in Schwarzschild and Rindler metrics. *Journal of Physics A: Mathematical and General*, 8(4):609, apr 1975. doi: 10.1088/0305-4470/8/4/022. URL <https://dx.doi.org/10.1088/0305-4470/8/4/022>.

- [44] Martin Davis. A Computer Program for Presburger’s Algorithm. *Symbolic Computation Automation of Reasoning* 1, 1957.
- [45] Mykola Dedushenko. Snowmass white paper: The quest to define QFT. *International Journal of Modern Physics A*, 38(04n05), February 2023. ISSN 1793-656X. doi: 10.1142/S0217751X23300028. URL <http://dx.doi.org/10.1142/S0217751X23300028>.
- [46] DeepMind. alphafold. <https://github.com/google-deepmind/alphafold>, 2021.
- [47] DeepSeek. 3FS. <https://github.com/deepseek-ai/3FS>, 2025.
- [48] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2019.
- [49] Kefan Dong and Tengyu Ma. STP: Self-play LLM Theorem Provers with Iterative Conjecturing and Proving, 2025. URL <https://arxiv.org/abs/2502.00212>.
- [50] X. Fan, T.G. Myers, B.A.D. Sukra, and G. Gabrielse. Measurement of the Electron Magnetic Moment. *Physical Review Letters*, 130(7), February 2023. ISSN 1079-7114. doi: 10.1103/physrevlett.130.071801. URL <http://dx.doi.org/10.1103/PhysRevLett.130.071801>.
- [51] Deborah Ferreira and André Freitas. Natural Language Premise Selection: Finding Supporting Statements for Mathematical Text. In *Proceedings of the Language Resources and Evaluation Conference*, 2020.
- [52] R. P. Feynman. Space-Time Approach to Quantum Electrodynamics. *Phys. Rev.*, 76:769–789, Sep 1949. doi: 10.1103/PhysRev.76.769. URL <https://link.aps.org/doi/10.1103/PhysRev.76.769>.
- [53] Vlad Firoiu, Eser Aygun, Ankit Anand, Zafarali Ahmed, Xavier Glorot, Laurent Orseau, Lei Zhang, Doina Precup, and Shibl Mourad. Training a First-Order Theorem Prover from Synthetic Data. In *International Conference on Learning Representations Workshop on Mathematical Reasoning in General Artificial Intelligence*, 2021.
- [54] Emily First, Markus Rabe, Talia Ringer, and Yuriy Brun. Baldur: Whole-Proof Generation and Repair with Large Language Models. In *Proceedings of the ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, 2023.
- [55] Klaus Fredenhagen. *An Introduction to Algebraic Quantum Field Theory*, pages 1–30. 2015. doi: 10.1007/978-3-319-21353-8_1.
- [56] Stephen A. Fulling. Nonuniqueness of Canonical Field Quantization in Riemannian Space-Time. *Phys. Rev. D*, 7:2850–2862, May 1973. doi: 10.1103/PhysRevD.7.2850. URL <https://link.aps.org/doi/10.1103/PhysRevD.7.2850>.
- [57] Aurélien Garivier and Eric Moulines. On upper-confidence bound policies for switching bandit problems. In *Proceedings of the 22nd International Conference on Algorithmic Learning Theory, ALT’11*, page 174–188, Berlin, Heidelberg, 2011. Springer-Verlag. ISBN 9783642244117.
- [58] Israel Gelfand and Mark Neumark. On the imbedding of normed rings into the ring of operators in Hilbert space. *Математический сборник*, 12(2):197–217, 1943.

- [59] Elliot Glazer, Ege Erdil, Tamay Besiroglu, Diego Chicharro, Evan Chen, Alex Gunning, Caroline Falkman Olsson, Jean-Stanislas Denain, Anson Ho, Emily de Oliveira Santos, Olli Järvinemi, Matthew Barnett, Robert Sandler, Matej Vrzala, Jaime Sevilla, Qiuyu Ren, Elizabeth Pratt, Lionel Levine, Grant Barkley, Natalie Stewart, Bogdan Grechuk, Tetiana Grechuk, Shreepranav Varma Enugandla, and Mark Wildon. *FrontierMath: A Benchmark for Evaluating Advanced Mathematical Reasoning in AI*, 2024. URL <https://arxiv.org/abs/2411.04872>.
- [60] Fabian Gloeckle, Baptiste Roziere, Amaury Hayat, and Gabriel Synnaeve. Temperature-Scaled Large Language Models for Lean Proofstep Prediction. In *International Conference on Neural Information Processing Systems Workshop on MATH-AI*, 2023.
- [61] Zarathustra A Goertzel, Karel Chvalovský, Jan Jakubův, Miroslav Olšák, and Josef Urban. Fast and Slow ENIGMAs and Parental Guidance. In *Proceedings of the International Symposium on Frontiers of Combining Systems*, 2021.
- [62] Zarathustra A Goertzel, Jan Jakubův, Cezary Kaliszyk, Miroslav Olšák, Jelle Piepenbrock, and Josef Urban. The Isabelle ENIGMA. In *Proceedings of the International Conference on Interactive Theorem Proving*, 2022.
- [63] Rudolf Haag. *Local Quantum Physics: Fields, Particles, Algebras*. Springer Science & Business Media, 2012.
- [64] Rudolf Haag and Daniel Kastler. An Algebraic Approach to Quantum Field Theory. *Journal of Mathematical Physics*, 5(7):848–861, 07 1964. ISSN 0022-2488. doi: 10.1063/1.1704187. URL <https://doi.org/10.1063/1.1704187>.
- [65] Brian C Hall. *Quantum Theory for Mathematicians*. Springer, 2013.
- [66] Jesse Michael Han, Jason Rute, Yuhuai Wu, Edward W Ayers, and Stanislas Polu. Proof Artifact Co-Training for Theorem Proving with Language Models. In *Proceedings of the International Conference on Learning Representations*, 2022.
- [67] Xiaotian Han, Yiren Jian, Xuefeng Hu, Haogeng Liu, Yiqi Wang, Qihang Fan, Yuang Ai, Huaibo Huang, Ran He, Zhenheng Yang, and Quanzeng You. Infimm-webmath-40b: Advancing multimodal pre-training for enhanced mathematical reasoning, 2024. URL <https://arxiv.org/abs/2409.12568>.
- [68] Thomas Little Heath et al. *The thirteen books of Euclid’s Elements*. Courier Corporation, 1956.
- [69] Werner Heisenberg. Die “beobachtbaren Größen” in der Theorie der Elementarteilchen. *Zeitschrift für Physik*, 120, July 1943. doi: 10.1007/BF01329800. URL <https://doi.org/10.1007/BF01329800>.
- [70] HEPLean. HEPLean/PhysLean. <https://github.com/HEPLean/PhysLean>, 2025.
- [71] HepLean Community. PhysLean, 2025. URL <https://physlean.com>.
- [72] David Hilbert. Mathematical problems. *Bulletin of the American Mathematical Society*, 5:437–479, 1902. ISSN 0022-2488. doi: 10.1090/S0002-9904-1902-00923-3. URL <https://doi.org/10.1090/S0002-9904-1902-00923-3>.
- [73] Stefan Hollands and Roberto Longo. A New Proof of the QNEC, 2025. URL <https://arxiv.org/abs/2503.04651>.

- [74] Daniel Huang, Prafulla Dhariwal, Dawn Song, and Ilya Sutskever. GamePad: A Learning Environment for Theorem Proving. In *Proceedings of the International Conference on Learning Representations*, 2019.
- [75] iNSPIRE HEP. iNSPIRE HEP Conferences, 2025. URL <https://inspirehep.net/conferences>.
- [76] Geoffrey Irving, Christian Szegedy, Alexander A Alemi, Niklas Eén, François Chollet, and Josef Urban. DeepMath - Deep Sequence Models for Premise Selection. In *Proceedings of the International Conference on Neural Information Processing Systems*, 2016.
- [77] Jan Jakubův and Josef Urban. Hammering Mizar by Learning Clause Guidance. In *Proceedings of the International Conference on Interactive Theorem Proving*, 2019.
- [78] Jan Jakubův, Karel Chvalovský, Miroslav Olšák, Bartosz Piotrowski, Martin Suda, and Josef Urban. ENIGMA Anonymous: Symbol-Independent Inference Guiding Machine (System Description). In *Proceedings of the International Joint Conference on Automated Reasoning*, 2020.
- [79] Jan Jakubův, Karel Chvalovský, Zarathustra A Goertzel, Cezary Kaliszyk, Mirek Olšák, Bartosz Piotrowski, Stephan Schulz, Martin Suda, and Josef Urban. MizAR 60 for Mizar 50. In *Proceedings of the International Conference on Interactive Theorem Proving*, 2023.
- [80] Albert Q Jiang, Wenda Li, Jesse Michael Han, and Yuhuai Wu. LISA: Language Models of Isabelle Proofs. In *Proceedings of the Conference on Artificial Intelligence and Theorem Proving*, 2021.
- [81] Albert Q Jiang, Wenda Li, and Mateja Jamnik. Multilingual Mathematical Autoformalization. *arXiv preprint arXiv:2311.03755*, 2023.
- [82] Albert Q. Jiang, Sean Welleck, Jin Peng Zhou, Wenda Li, Jiacheng Liu, Mateja Jamnik, Timothée Lacroix, Yuhuai Wu, and Guillaume Lample. Draft, Sketch, and Prove: Guiding Formal Theorem Provers with Informal Proofs, 2023. URL <https://arxiv.org/abs/2210.12283>.
- [83] Res Jost. Eine Bemerkung zum CTP theorem. *Helv. Phys. Acta*, 30(409):153, 1957.
- [84] Cezary Kaliszyk, François Chollet, and Christian Szegedy. HolStep: A Machine Learning Dataset for Higher-Order Logic Theorem Proving. In *Proceedings of the International Conference on Learning Representations*, 2017.
- [85] Guillaume Lample, Timothee Lacroix, Marie-Anne Lachaux, Aurelien Rodriguez, Amaury Hayat, Thibaut Lavril, Gabriel Ebner, and Xavier Martinet. Hypertree Proof Search for Neural Theorem Proving. In *Proceedings of the International Conference on Neural Information Processing Systems*, 2022.
- [86] Tor Lattimore and Csaba Szepesvári. *Bandit Algorithms*. Cambridge University Press, 2020.
- [87] Lean Community. Creating a Lean project. <https://leanprover-community.github.io/install/project.html#creating-a-lean-project>, 2025.
- [88] Lean Community. Lean-related conferences and events. <https://leanprover-community.github.io/events.html>, 2025.
- [89] Lean Community. Lean Zulip Server, 2025. URL <https://leanprover.zulipchat.com/#narrow/channel/113486-announce>.
- [90] H. Lehmann, K. Symanzik, and W. Zimmermann. On the formulation of quantized field theories. *Nuovo Cim.*, 1:205–225, 1955. doi: 10.1007/BF02731765.

- [91] Leni Aniva, Chuyue Sun, Brando Miranda, Clark Barrett, and Sanmi Koyejo. stanford-centaur/PyPantograph. <https://github.com/stanford-centaur/PyPantograph>, 2025.
- [92] Robert Y. Lewis. A formal proof of Hensel’s lemma over the p-adic integers. In *Proceedings of the 8th ACM SIGPLAN International Conference on Certified Programs and Proofs, CPP ’19*, page 15–26. ACM, January 2019. doi: 10.1145/3293880.3294089. URL <http://dx.doi.org/10.1145/3293880.3294089>.
- [93] Zhaoyu Li, Jialiang Sun, Logan Murphy, Qidong Su, Zenan Li, Xian Zhang, Kaiyu Yang, and Xujie Si. A Survey on Deep Learning for Theorem Proving, 2024. URL <https://arxiv.org/abs/2404.09939>.
- [94] Yong Lin, Shange Tang, Bohan Lyu, Jiayun Wu, Hongzhou Lin, Kaiyu Yang, Jia Li, Mengzhou Xia, Danqi Chen, Sanjeev Arora, and Chi Jin. Goedel-Prover: A Frontier Model for Open-Source Automated Theorem Proving, 2025. URL <https://arxiv.org/abs/2502.07640>.
- [95] Chengwu Liu, Jianhao Shen, Huajian Xin, Zhengying Liu, Ye Yuan, Haiming Wang, Wei Ju, Chuanyang Zheng, Yichun Yin, Lin Li, et al. FIMO: A Challenge Formal Dataset for Automated Theorem Proving. *arXiv preprint arXiv:2309.04295*, 2023.
- [96] Roberto Longo. A Bekenstein-type bound in QFT, 2024. URL <https://arxiv.org/abs/2409.14408>.
- [97] Roberto Longo and Edward Witten. A note on continuous entropy, 2022. URL <https://arxiv.org/abs/2202.03357>.
- [98] Sarah Loos, Geoffrey Irving, Christian Szegedy, and Cezary Kaliszyk. Deep Network Guided Proof Search. In *Proceedings of the International Conference on Logic for Programming, Artificial Intelligence and Reasoning*, 2017.
- [99] The mathlib Community. The Lean Mathematical Library. In *Proceedings of the ACM SIGPLAN International Conference on Certified Programs and Proofs*, 2020.
- [100] Vincenzo Morinelli. An algebraic condition for the Bisognano-Wichmann Property, 2016. URL <https://arxiv.org/abs/1604.04750>.
- [101] Leonardo de Moura and Sebastian Ullrich. The Lean 4 Theorem Prover and Programming Language. In André Platzer and Geoff Sutcliffe, editors, *Automated Deduction – CADE 28*, pages 625–635, Cham, 2021. Springer International Publishing. ISBN 978-3-030-79876-5.
- [102] Mozilla. mozilla/TTS. <https://github.com/mozilla/TTS>, 2025.
- [103] Mozilla. mozilla/DeepSpeech. <https://github.com/mozilla/DeepSpeech>, 2025.
- [104] Nat Friedman. Introducing GitHub Copilot: your AI pair programmer, 2021. URL <https://github.blog/news-insights/product-news/introducing-github-copilot-ai-pair-programmer/>.
- [105] Yazhe Niu, Yuan Pu, Zhenjie Yang, Xueyan Li, Tong Zhou, Jiyuan Ren, Shuai Hu, Hongsheng Li, and Yu Liu. LightZero: A Unified Benchmark for Monte Carlo Tree Search in General Sequential Decision Scenarios. *Advances in Neural Information Processing Systems*, 36, 2024.
- [106] Norwegian Academy of Science and Letters. Michel Talagrand awarded the 2024 Abel Prize, 2024. URL <https://abelprize.no/article/2024/michel-talagrand-awarded-2024-abel-prize>.
- [107] OpenAI. Introducing ChatGPT, 2022. URL <https://openai.com/index/chatgpt/>.

- [108] Konrad Osterwalder and Robert Schrader. Axioms for Euclidean Green’s functions. *Commun. Math. Phys.*, 31:83–112, 1973. doi: 10.1007/BF01645738.
- [109] Konrad Osterwalder and Robert Schrader. Axioms for Euclidean Green’s functions. *Communications in Mathematical Physics*, 31(2):83–112, 1973.
- [110] Konrad Osterwalder and Robert Schrader. Axioms for Euclidean Green’s Functions 2. *Commun. Math. Phys.*, 42:281, 1975. doi: 10.1007/BF01608978.
- [111] Aditya Paliwal, Sarah Loos, Markus Rabe, Kshitij Bansal, and Christian Szegedy. Graph Representations for Higher-Order Logic and Theorem Proving. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020.
- [112] Papers With Code. AI Conference Deadlines, 2025. URL <https://aideadlin.es/?sub=ML>.
- [113] Papers with Code. Automated Theorem Proving on miniF2F-test, 2025. URL <https://paperswithcode.com/sota/automated-theorem-proving-on-minif2f-test>.
- [114] Patrick Massot. PatrickMassot/leanblueprint. <https://github.com/PatrickMassot/leanblueprint>, 2025.
- [115] Pietro Monticone. pitmonticone/LeanProject. <https://github.com/pitmonticone/LeanProject>, 2025.
- [116] Stanislas Polu and Ilya Sutskever. Generative Language Modeling for Automated Theorem Proving. *arXiv preprint arXiv:2009.03393*, 2020.
- [117] Stanislas Polu, Jesse Michael Han, Kunhao Zheng, Mantas Baksys, Igor Babuschkin, and Ilya Sutskever. Formal Mathematics Statement Curriculum Learning. In *Proceedings of the International Conference on Learning Representations*, 2023.
- [118] Michael Rawson and Giles Reger. A Neurally-Guided, Parallel Theorem Prover. In *Proceedings of the International Symposium on Frontiers of Combining Systems*, 2019.
- [119] Z. Z. Ren, Zhihong Shao, Junxiao Song, Huajian Xin, Haocheng Wang, Wanjia Zhao, Liyue Zhang, Zhe Fu, Qihao Zhu, Dejian Yang, Z. F. Wu, Zhibin Gou, Shirong Ma, Hongxuan Tang, Yuxuan Liu, Wenjun Gao, Daya Guo, and Chong Ruan. DeepSeek-Prover-V2: Advancing Formal Mathematical Reasoning via Reinforcement Learning for Subgoal Decomposition, 2025. URL <https://arxiv.org/abs/2504.21801>.
- [120] Eckhardt Roger. Stan Ulam, John von Neumann, and the Monte Carlo Method. *Los Alamos Science*, 15:131–137, 1987.
- [121] Jason Rute, Miroslav Olšák, Lasse Blaauwbroek, Fidel Ivan Schaposnik Massolo, Jelle Piepenbrock, and Vasily Pestun. Graph2Tac: Online Representation Learning of Formal Math Concepts. In *Proceedings of the International Conference on Machine Learning*, 2024.
- [122] Alex Sanchez-Stern, Yousef Alhessi, Lawrence Saul, and Sorin Lerner. Generating Correctness Proofs with Neural Networks. In *Proceedings of the ACM SIGPLAN International Workshop on Machine Learning and Programming Languages*, 2020.
- [123] Marta Sanz-Solé, Javier Soria, Juan Luis Varona, and Joan Verdera. Proceedings of the International Congress of Mathematicians 2006. Volume I, 2006. URL <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf>.

- [124] I. E. Segal. Irreducible representations of operator algebras. *Bull. Amer. Math. Soc.*, 53(12):73–88, 1947. URL <http://dml.mathdoc.fr/item/1183510397>.
- [125] Irving E Segal. Postulates for general quantum mechanics. *Annals of Mathematics*, 48(4):930–948, 1947.
- [126] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models, 2024. URL <https://arxiv.org/abs/2402.03300>.
- [127] David Silver, Aja Huang, Chris J. Maddison, Arthur Guez, Laurent Sifre, George van den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, Marc Lanctot, Sander Dieleman, Dominik Grewe, John Nham, Nal Kalchbrenner, Ilya Sutskever, Timothy Lillicrap, Madeleine Leach, Koray Kavukcuoglu, Thore Graepel, and Demis Hassabis. Mastering the game of go with deep neural networks and tree search. *Nature*, 529(7587):484–489, January 2016. doi: 10.1038/nature16961.
- [128] David Silver, Julian Schrittwieser, Karen Simonyan, Ioannis Antonoglou, Aja Huang, Arthur Guez, Thomas Hubert, Lucas baker, Matthew Lai, Adrian Bolton, Yutian Chen, Timothy P. Lillicrap, Fan Hui, L. Sifre, George van den Driessche, Thore Graepel, and Demis Hassabis. Mastering the game of Go without human knowledge. *Nature*, 550:354–359, 2017. URL <https://api.semanticscholar.org/CorpusID:205261034>.
- [129] David Silver, Thomas Hubert, Julian Schrittwieser, Ioannis Antonoglou, Matthew Lai, Arthur Guez, Marc Lanctot, Laurent Sifre, Dhharshan Kumaran, Thore Graepel, et al. A General Reinforcement Learning Algorithm That Masters Chess, Shogi, and Go Through Self-Play. *Science*, 2018.
- [130] Peiyang Song, Kaiyu Yang, and Anima Anandkumar. lean-doj/leancopilot. <https://github.com/lean-doj/LeanCopilot>, 2025.
- [131] Peiyang Song, Kaiyu Yang, and Anima Anandkumar. Lean Copilot: Large Language Models as Copilots for Theorem Proving in Lean, 2025. URL <https://arxiv.org/abs/2404.12534>.
- [132] Jonathan Sorce. A short proof of Tomita’s theorem. *Journal of Functional Analysis*, 286(12):110420, June 2024. ISSN 0022-1236. doi: 10.1016/j.jfa.2024.110420. URL <http://dx.doi.org/10.1016/j.jfa.2024.110420>.
- [133] Raymond F. Streater and Arthur S. Wightman. *PCT, Spin and Statistics, and All That*. Princeton University Press, 1989. URL <http://www.jstor.org/stable/j.ctt1cx3vcq>.
- [134] Alexander Strohmaier and Edward Witten. The Timelike Tube Theorem in Curved Spacetime, 2023. URL <https://arxiv.org/abs/2303.16380>.
- [135] Alexander Strohmaier and Edward Witten. Analytic states in quantum field theory on curved spacetimes, 2024. URL <https://arxiv.org/abs/2302.02709>.
- [136] Martin Suda. Vampire with a Brain Is a Good ITP Hammer. In *Proceedings of the International Symposium on Frontiers of Combining Systems*, 2021.
- [137] Martin Suda. Improving ENIGMA-Style Clause Selection While Learning From History. In *Proceedings of the International Conference on Automated Deduction*, 2021.

- [138] Stephen J. Summers. Tomita-Takesaki Modular Theory, 2005. URL <https://arxiv.org/abs/math-ph/0511034>.
- [139] Ilya Sutskever, Oriol Vinyals, and Quoc V Le. Sequence to Sequence Learning with Neural Networks. In *Proceedings of the International Conference on Neural Information Processing Systems*, 2014.
- [140] Christian Suttner and Wolfgang Ertel. Automatic Acquisition of Search Guiding Heuristics. In *Proceedings of the International Conference on Automated Deduction*, 1990.
- [141] Zirô Takeda. Inductive limit and infinite direct product of operator algebras. *Tohoku Mathematical Journal, Second Series*, 7(1-2):67–86, 1955.
- [142] M. Takesaki. *Tomita's Theory of Modular Hilbert Algebras and its Applications*. Lecture Notes in Mathematics. Springer-Verlag, 1970. doi: 10.1007/bfb0065832.
- [143] Michel Talagrand. *What Is a Quantum Field Theory?* Cambridge University Press, 2022.
- [144] Terence Tao. Terence Tao at IMO 2024: AI and Mathematics, 2024. URL <https://youtu.be/e049IoFBnLA?feature=shared&t=3005>.
- [145] Masamichi Tekesaki. Lecture 01 by Masamichi Tekesaki, 2014. URL https://youtu.be/23-Aumh91_c?feature=shared.
- [146] The Royal Swedish Academy of Sciences. Press release: The Nobel Prize in Chemistry 2024, 2024. URL <https://www.nobelprize.org/prizes/chemistry/2024/press-release/>.
- [147] Minoru Tomita. On canonical forms of von Neumann algebras. *Fifth Functional Analysis Sympos (Tôhoku Univ., Sendai, 1967)*, *Math. Inst., Tohoku Univ., Sendai*, pages 101–102, 1967.
- [148] Joseph Tooby-Smith. HepLean: Digitalising high energy physics, 2024. URL <https://arxiv.org/abs/2405.08863>.
- [149] Trieu H Trinh, Yuhuai Wu, Quoc V Le, He He, and Thang Luong. Solving Olympiad Geometry without Human Demonstrations. *Nature*, 2024.
- [150] W. G. Unruh. Notes on black-hole evaporation. *Phys. Rev. D*, 14:870–892, Aug 1976. doi: 10.1103/PhysRevD.14.870. URL <https://link.aps.org/doi/10.1103/PhysRevD.14.870>.
- [151] Rahul Vishwakarma and Subhankar Mishra. Enhancing Neural Theorem Proving through Data Augmentation and Dynamic Sampling Method. *arXiv preprint arXiv:2312.14188*, 2023.
- [152] Robert M Wald. Quantum field theory in curved spacetime. *arXiv preprint gr-qc/9509057*, 1995.
- [153] Haiming Wang, Ye Yuan, Zhengying Liu, Jianhao Shen, Yichun Yin, Jing Xiong, Enze Xie, Han Shi, Yujun Li, Lin Li, et al. DT-Solver: Automated Theorem Proving with Dynamic-Tree Sampling Guided by Proof-level Value Function. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, 2023.
- [154] Haiming Wang, Mert Unsal, Xiaohan Lin, Mantas Baksys, Junqi Liu, Marco Dos Santos, Flood Sung, Marina Vinyes, Zhenzhe Ying, Zekai Zhu, Jianqiao Lu, Hugues de Saxcé, Bolton Bailey, Chendong Song, Chenjun Xiao, Dehao Zhang, Ebony Zhang, Frederick Pu, Han Zhu, Jiawei Liu, Jonas Bayer, Julien Michel, Longhui Yu, Léo Dreyfus-Schmidt, Lewis Tunstall, Luigi Pagani, Moreira Machado, Pauline Bourigault, Ran Wang, Stanislas Polu, Thibaut Barroyer, Wen-Ding Li,

- Yazhe Niu, Yann Fleureau, Yangyang Hu, Zhouliang Yu, Zihan Wang, Zhilin Yang, Zhengying Liu, and Jia Li. Kimina-Prover Preview: Towards Large Formal Reasoning Models with Reinforcement Learning, 2025. URL <https://arxiv.org/abs/2504.11354>.
- [155] Qingxiang Wang, Cezary Kaliszyk, and Josef Urban. First Experiments with Neural Translation of Informal to Formal Mathematics. In *Proceedings of the International Conference on Intelligent Computer Mathematics*, 2018.
- [156] Qingxiang Wang, Chad Brown, Cezary Kaliszyk, and Josef Urban. Exploration of Neural Machine Translation in Autoformalization of Mathematics in Mizar. In *Proceedings of the ACM SIGPLAN International Conference on Certified Programs and Proofs*, 2020.
- [157] Chenrui Wei, Mengzhou Sun, and Wei Wang. Proving Olympiad Algebraic Inequalities without Human Demonstrations. *arXiv preprint arXiv:2406.14219*, 2024.
- [158] Steven Weinberg. *The Quantum theory of fields. Vol. 1: Foundations*. Cambridge University Press, 6 2005. ISBN 978-0-521-67053-1, 978-0-511-25204-4. doi: 10.1017/CBO9781139644167.
- [159] Steven Weinberg. *The quantum theory of fields. Vol. 2: Modern applications*. Cambridge University Press, 8 2013. ISBN 978-1-139-63247-8, 978-0-521-67054-8, 978-0-521-55002-4. doi: 10.1017/CBO9781139644174.
- [160] Sean Welleck and Rahul Saha. LLMSTEP: LLM Proofstep Suggestions in Lean. In *International Conference on Neural Information Processing Systems Workshop on MATH-AI*, 2023.
- [161] Sean Welleck, Jiacheng Liu, Ronan Le Bras, Hannaneh Hajishirzi, Yejin Choi, and Kyunghyun Cho. NaturalProofs: Mathematical Theorem Proving in Natural Language. In *Proceedings of the International Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2021.
- [162] Daniel Whalen. Holophrasm: A Neural Automated Theorem Prover for Higher-Order Logic. *arXiv preprint arXiv:1608.02644*, 2016.
- [163] Wikipedia Community. Lee Sedol, 2025. URL https://en.wikipedia.org/wiki/Lee_Sedol.
- [164] Edward Witten. APS Medal for Exceptional Achievement in Research: Invited article on entanglement properties of quantum field theory. *Reviews of Modern Physics*, 90(4), October 2018. ISSN 1539-0756. doi: 10.1103/revmodphys.90.045003. URL <http://dx.doi.org/10.1103/RevModPhys.90.045003>.
- [165] Edward Witten. Algebras, Regions, and Observers, 2023. URL <https://arxiv.org/abs/2303.02837>.
- [166] Edward Witten. A Background Independent Algebra in Quantum Gravity, 2023. URL <https://arxiv.org/abs/2308.03663>.
- [167] Yuhuai Wu, Albert Q Jiang, Jimmy Ba, and Roger Grosse. INT: An Inequality Benchmark for Evaluating Generalization in Theorem Proving. In *Proceedings of the International Conference on Learning Representations*, 2021.
- [168] Yuhuai Wu, Markus N Rabe, Wenda Li, Jimmy Ba, Roger B Grosse, and Christian Szegedy. LIME: Learning Inductive Bias for Primitives of Mathematical Reasoning. In *Proceedings of the International Conference on Machine Learning*, 2021.

- [169] Yuhuai Wu, Albert Q Jiang, Wenda Li, Markus Rabe, Charles Staats, Mateja Jamnik, and Christian Szegedy. Autoformalization with Large Language Models. In *Proceedings of the International Conference on Neural Information Processing Systems*, 2022.
- [170] Zijian Wu, Suozhi Huang, Zhejian Zhou, Huaiyuan Ying, Jiayu Wang, Dahua Lin, and Kai Chen. InternLM2.5-StepProver: Advancing Automated Theorem Proving via Expert Iteration on Large-Scale LEAN Problems, 2024. URL <https://arxiv.org/abs/2410.15700>.
- [171] Huajian Xin, Z. Z. Ren, Junxiao Song, Zhihong Shao, Wanbiao Zhao, Haocheng Wang, Bo Liu, Liyue Zhang, Xuan Lu, Qiushi Du, Wenjun Gao, Qihao Zhu, Dejian Yang, Zhibin Gou, Z. F. Wu, Fuli Luo, and Chong Ruan. DeepSeek-Prover-V1.5: Harnessing Proof Assistant Feedback for Reinforcement Learning and Monte-Carlo Tree Search, 2024. URL <https://arxiv.org/abs/2408.08152>.
- [172] Huajian Xin, Haiming Wang, Chuanyang Zheng, Lin Li, Zhengying Liu, Qingxing Cao, Yinya Huang, Jing Xiong, Han Shi, Enze Xie, et al. LEGO-Prover: Neural Theorem Proving with Growing Libraries. In *Proceedings of the International Conference on Learning Representations*, 2024.
- [173] Jing Xiong, Jianhao Shen, Ye Yuan, Haiming Wang, Yichun Yin, Zhengying Liu, Lin Li, Zhijiang Guo, Qingxing Cao, Yinya Huang, et al. TRIGO: Benchmarking Formal Mathematical Proof Reduction for Generative Language Models. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 2023.
- [174] Kaiyu Yang, Aidan Swope, Alex Gu, Rahul Chalamala, Peiyang Song, Shixing Yu, Saad Godil, Ryan Prenger, and Anima Anandkumar. LeanDojo: Theorem Proving with Retrieval-Augmented Language Models. In *Proceedings of the International Conference on Neural Information Processing Systems*, 2023.
- [175] Eric Yeh, Briland Hitaj, Sam Owre, Maena Quemener, and Natarajan Shankar. CoProver: A Recommender System for Proof Construction. In *Proceedings of the International Conference on Intelligent Computer Mathematics*, 2023.
- [176] Huaiyuan Ying, Shuo Zhang, Linyang Li, Zhejian Zhou, Yunfan Shao, Zhaoye Fei, Yichuan Ma, Jiawei Hong, Kuikun Liu, Ziyi Wang, et al. InternLM-Math: Open Math Large Language Models Toward Verifiable Reasoning. *arXiv preprint arXiv:2402.06332*, 2024.
- [177] Liao Zhang, Lasse Blaauwbroek, Cezary Kaliszyk, and Josef Urban. Learning Proof Transformations and Its Applications in Interactive Theorem Proving. In *Proceedings of the International Symposium on Frontiers of Combining Systems*, 2023.
- [178] Shizhuo Dylan Zhang, Talia Ringer, and Emily First. Getting More out of Large Language Models for Proofs. In *Proceedings of the Conference on Artificial Intelligence and Theorem Proving*, 2023.
- [179] Kunhao Zheng, Jesse Michael Han, and Stanislas Polu. MiniF2F: A Cross-System Benchmark for Formal Olympiad-Level Mathematics. In *Proceedings of the International Conference on Learning Representations*, 2022.
- [180] Jin Peng Zhou, Yuhuai Wu, Qiyang Li, and Roger Grosse. REFACTOR: Learning to Extract Theorems from Proofs. In *Proceedings of the International Conference on Learning Representations*, 2024.
- [181] Zsolt Zombori, Josef Urban, and Miroslav Olšák. The Role of Entropy in Guiding a Connection Prover. In *Proceedings of the International Conference on Automated Reasoning with Analytic Tableaux and Related Methods*, 2021.